Lesson 1 — Extra Practice

Multiplying and Factoring — The Reverse Relationship

Part A — Round Trips: Expand, Factor, and Check

Expanding and factoring are the same road travelled in opposite directions. If you expand a product and then factor the result, you must arrive back where you started. Every question in this part ends with a **check** — compare your answer to the original expression, by expanding or by substituting a number into both forms.

1. Expand each product. Then check your expansion by substituting $x = 2$ into both the product and your expanded answer — the two values must match.

(a) $(x + 3)(x + 7)$

(b) $(x + 2)(x + 9)$

(c) $(x - 4)(x - 5)$

(d) $(x + 6)(x - 2)$

2. Expand each product, then factor the result **completely**. In one of these four, the starting form was not completely factored — find it.

(a) $5(2x + 3)$ (b) $3x(x - 4)$ (c) $4x(2x + 7)$ (d) $6(x^2 - 3x)$

3. Now run the trip the other way. Factor each expression, then expand your factored form to verify it reproduces the original exactly.

(a) $8x^2 + 12x$ (b) $6x^2 - 9x$ (c) $10x^3 + 15x^2$

4. Each product below has **three** factors. Expand it completely, then check your answer by substituting $x = 1$ into both the product and your expanded form.

(a) $2(x + 1)(x + 5)$

(b) $4(x + 2)(x - 3)$

Part B — Checking a Factorization by Expanding

A proposed factorization is worth nothing until it is checked. Expanding is the check: multiply the factors out and compare, term by term, with the original.

5. Expand the proposed factored form to decide whether it is correct. Write **correct** or **incorrect**; if it is incorrect, write what the proposed form really expands to.

(a) Is $x^2 + 8x + 12 = (x + 2)(x + 6)$? (b) Is $x^2 - 5x + 6 = (x - 2)(x - 3)$? (c) Is $x^2 + 13x + 12 = (x + 3)(x + 4)$?

(d) Is $x^2 - x - 20 = (x - 5)(x + 4)$? (e) Is $x^2 + 2x - 8 = (x + 2)(x - 4)$? (f) Is $x^2 - 9x + 18 = (x - 3)(x - 6)$?

6. Expand to check each proposed factoring. One of them expands back correctly but is still **not completely factored** — identify it and finish the job.

(a) Is $12x + 18 = 2(6x + 9)$?

(b) Is $10x^2 - 15x = 5x(2x - 3)$?

(c) Is $14x + 21 = 7(2x + 4)$?

(d) Is $4x^3 + 6x^2 = 2x^2(2x + 3)$?

7. **Error analysis.** A student claims that $x^2 + 9 = (x + 3)(x + 3)$, reasoning that "$3 \times 3 = 9$." Expand $(x + 3)(x + 3)$ to disprove the claim, and explain in one sentence which term the student forgot about. Then check the claim a second way by substituting $x = 1$ into both sides.

8. **Substitution as a check.** A factored form and an expanded form describe the same expression, so they must give the same value at *every* value of x.

(a) Verify that $x^2 + 8x + 12$ and $(x + 2)(x + 6)$ agree by substituting $x = 1$ and then $x = 3$ into both forms.

(b) Substitute $x = 0$ into x^2+2x-8 and into $(x+2)(x-4)$. The values match — yet the factoring is wrong. Now substitute $x = 1$ into both, and explain what this shows about testing a single value.

Part C — Sorting and Matching

9. Match each **expanded** expression in the left column with its **factored** form in the scrambled right column. Write the pairs, such as "A — VI".

A. $x^2 + 9x + 18$	I. $(x - 6)(x + 3)$
B. $x^2 + 3x - 18$	II. $(x + 6)(x + 3)$
C. $x^2 - 9x + 18$	III. $(x - 6)(x - 3)$
D. $x^2 - 3x - 18$	IV. $(x + 6)(x - 3)$
E. $2x^2 + 12x$	V. $(x - 6)(x + 6)$
F. $x^2 - 36$	VI. $2x(x + 6)$

10. Sort each expression as **expanded**, **factored**, or **neither**. An expression is "neither" when it is written as a sum whose terms are not all fully multiplied out, or as a product that is not finished. Give a one-line reason for each "neither".

(a) $(x + 2)(x + 9)$ (b) $x^2 + 11x + 18$ (c) $2(x^2 + 3x)$ (d) $3x(x + 5)$

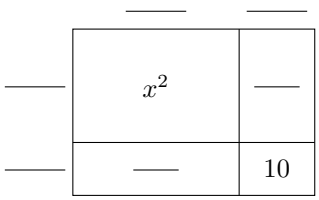
(e) $3x^2 + 15x$ (f) $x(x + 4) + 2(x + 4)$ (g) $5(x + 1)(x + 4)$ (h) $x^2 - 49$

11. Each expression below is a product, so it looks factored — but only some are **completely** factored. Decide which, and rewrite the others in completely factored form.

(a) $2(2x + 6)$ (b) $3x(x + 2)$ (c) $x(3x + 9)$ (d) $5(x^2 + 4x)$

Part D — Area Models, Working Backwards, and Reasoning

12. **Area model.** A rectangle has an area of $x^2 + 7x + 10$ square units. The diagram shows the rectangle cut into four smaller rectangles, one for each term of the expansion.



(a) Fill in the two missing inside areas and the four side lengths. What are the length and width of the whole rectangle?

(b) Write the area as a product of the two side lengths, then expand that product to confirm you get $x^2 + 7x + 10$ back.

13. Each rectangle's area and one side length are given. Find the other side length by factoring, then check by multiplying the two sides together.

(a) Area $= 6x^2 + 15x$, width $= 3x$

(b) Area $= 12x + 8$, width $= 4$

(c) Area $= x^2 + 7x$, width $= x$

14. **Working backwards.** Suppose $ax(x + b) = 5x^2 + 35x$, where a and b are positive whole numbers.

(a) Expand $ax(x+b)$ and explain what a and $a \times b$ must equal.

(b) Use part (a) to find a and b.

(c) Write the factored form, then expand it to verify that you get $5x^2+35x$ back.

15. Fill in each blank so that the statement is true. Expand to check every answer.

(a) $(x + 3)(x + ____)$ $= x^2 + ____x + 21$

(b) $(x - 5)(x + ____)$ $= x^2 + ____x - 40$

(c) $(x + ____)(x + 4)$ $= x^2 + 11x + ____$

(d) $_____(x + 4)$ $= 6x + 24$

16. **Reasoning.** Answer each in a sentence or two.

(a) Expanding a product is a straightforward procedure, but factoring often requires searching. Explain why, using $(x + 4)(x + 5)$ and $x^2 + 9x + 20$ as your two examples.

(b) What does it mean for a factorization to be **complete**? Use $12x + 18 = 2(6x + 9)$ to explain why an answer can expand back correctly and still be unfinished.

17. Ravi says, "Every polynomial can be factored, because you can always write it as 1 times itself." Explain why writing $x^2 + 5x + 3 = 1(x^2 + 5x + 3)$ is not considered a useful factoring, and what we normally require of the factors instead.

18. **Stretch.** In the expression $12x^2 + kx$, the number k is a positive whole number less than 30.

(a) List *every* value of k for which the GCF of $12x^2 + kx$ is exactly $4x$.

(b) For the largest k on your list, write $12x^2 + kx$ in factored form and expand to check.

(c) Explain why $k = 24$ is *not* on your list, and give the GCF of $12x^2 + 24x$.

Answer Key. 1. (a) $x^2+10x+21$; at $x=2$: $5\times9=45$ and $4+20+21=45$ ✓ (b) $x^2+11x+18$; $4\times11=44$ and $4+22+18=44$ ✓ (c) $x^2-9x+20$; $(-2)(-3)=6$ and $4-18+20=6$ ✓ (d) $x^2+4x-12$; $8\times0=0$ and $4+8-12=0$ ✓. **2.** (a) $10x+15=5(2x+3)$ (b) $3x^2-12x=3x(x-4)$ (c) $8x^2+28x=4x(2x+7)$ (d) $6x^2-18x=6x(x-3)$ — part (d) was the incomplete one, since x^2-3x still had a common factor of x. **3.** (a) $4x(2x+3)$; $4x(2x+3)=8x^2+12x$ ✓ (b) $3x(2x-3)$; $=6x^2-9x$ ✓ (c) $5x^2(2x+3)$; $=10x^3+15x^2$ ✓. **4.** (a) $2x^2+12x+10$; at $x=1$: $2(2)(6)=24$ and $2+12+10=24$ ✓ (b) $4x^2-4x-24$; $4(3)(-2)=-24$ and $4-4-24=-24$ ✓. **5.** (a) correct $(x^2+8x+12)$ (b) correct (x^2-5x+6) (c) incorrect — $(x+3)(x+4)=x^2+7x+12$, not $x^2+13x+12$ (d) correct (x^2-x-20) (e) incorrect — $(x+2)(x-4)=x^2-2x-8$, not x^2+2x-8 (f) correct $(x^2-9x+18)$. **6.** (a) expands correctly to $12x+18$, but is *not complete* — $6x+9$ still has a common factor of 3, so $12x+18=6(2x+3)$ (b) correct and complete (c) incorrect — $7(2x+4)=14x+28$; correct form $7(2x+3)$ (d) correct and complete. **7.** $(x+3)(x+3)=x^2+6x+9$, not x^2+9; the student ignored the two middle products $3x$ and $3x$. Substituting $x=1$: $x^2+9=10$ but $(1+3)(1+3)=16$, so the claim fails. (x^2+9 does not factor over the integers.) **8.** (a) $x=1$: $1+8+12=21$ and $(3)(7)=21$ ✓; $x=3$: $9+24+12=45$ and $(5)(9)=45$ ✓ (b) $x=0$: -8 and $(2)(-4)=-8$, a match; $x=1$: $1+2-8=-5$ but $(3)(-3)=-9$. Agreeing at one value proves nothing — a correct factoring must agree at *every* value, so a single mismatch is enough to disprove it. **9.** A — II, B — IV, C — III, D — I, E — VI, F — V. **10.** (a) factored (b) expanded (c) neither — the bracket x^2+3x still factors, so the product is unfinished (d) factored (e) expanded (f) neither — it is a sum of two products, not a single product; it becomes $(x+4)(x+2)$ (g) factored (h) expanded. **11.** (a) not complete; $2(2x+6)=4(x+3)$ (b) complete (c) not complete; $x(3x+9)=3x(x+3)$ (d) not complete; $5(x^2+4x)=5x(x+4)$. **12.** (a) Missing inside areas $5x$ (top right) and $2x$ (bottom left); side lengths x and 5 across the top, x and 2 down the side. The rectangle is $(x+5)$ by $(x+2)$ (b) $(x+2)(x+5)=x^2+7x+10$ ✓. **13.** (a) $6x^2+15x=3x(2x+5)$, so length $=2x+5$ (b) $12x+8=4(3x+2)$, so length $=3x+2$ (c) $x^2+7x=x(x+7)$, so length $=x+7$. **14.** (a) $ax(x+b)=ax^2+abx$, so $a=5$ and $a\times b=35$ (b) $a=5$, $b=7$ (c) $5x(x+7)=5x^2+35x$ ✓. **15.** (a) 7 and 10: $(x+3)(x+7)=x^2+10x+21$ (b) 8 and 3: $(x-5)(x+8)=x^2+3x-40$ (c) 7 and 28: $(x+7)(x+4)=x^2+11x+28$ (d) 6: $6(x+4)=6x+24$. **16.** (a) Expanding $(x+4)(x+5)$ is a fixed routine — multiply every term by every term and collect, and there is only one possible answer. Factoring $x^2+9x+20$ runs backwards: you must search for two numbers with product 20 and sum 9, testing candidates until a pair works, and sometimes none exists. (b) A factorization is complete when no factor can be broken down any further. $2(6x+9)$ does expand back to $12x+18$, but $6x+9=3(2x+3)$, so a common factor of 3 was left inside the bracket; the complete answer is $6(2x+3)$. **17.** 1 times the expression is always true but changes nothing — no factor has been broken off, so it gives no new information and cannot help solve or simplify. A useful factoring writes the expression as a product of factors of *lower degree* (or a numerical factor greater than 1), each of which cannot be factored further. **18.** (a) $k=4,\ 8,\ 16,\ 20,\ 28$ — the multiples of 4 that are not multiples of 3 (so the number part of the GCF is exactly 4) (b) $12x^2+28x=4x(3x+7)$; $4x(3x+7)=12x^2+28x$ ✓ (c) 24 is a multiple of 12, so the GCF is $12x$, not $4x$: $12x^2+24x=12x(x+2)$.

Lesson 2 — Extra Practice

Common Factors: Finding and Removing the GCF

Part A — Longer Polynomials, More Variables, Negative GCFs

1. State the greatest common factor of each set of monomials. Where every monomial is negative, follow the course convention and give a *negative* GCF.

(a) $24x^4y^2,\ 36x^3y^3,\ 60x^2y$

(b) $18a^5b,\ 27a^3b^4,\ 45a^2b^2$

(c) $-14m^3n^2,\ -21m^2n^3,\ -35m^4n$

(d) $30x^3,\ 45x^2y,\ 75xy^2$

2. Factor each **four-term** polynomial by removing the GCF. There is one more term to divide, but the routine is unchanged. Verify by expanding.

(a) $12x^4 - 18x^3 + 24x^2 - 6x$

(b) $20a^5 + 30a^4 - 50a^3 + 10a^2$

(c) $9y^4 - 27y^3 + 18y^2 - 45y$

3. Factor each polynomial. In every case the GCF contains **both** variables, each at its lowest exponent.

(a) $15a^3b^2 - 25a^2b^3 + 10a^2b^2$

(b) $24x^3y^2 - 36x^2y^3 + 12x^2y^2$

(c) $14m^4n^2 + 21m^3n^3 - 28m^2n^4$

4. Each leading term is negative, so remove a *negative* GCF and leave a positive leading term inside the bracket.

(a) $-8x^3 + 12x^2 - 20x$

(b) $-18a^4 - 24a^3 + 30a^2$

(c) $-12x^3y + 18x^2y^2 - 30xy^3$

(d) $-16m^5 + 40m^3 - 24m^2$

Part B — Working Backwards, and the Vanishing 1

5. Fill in each blank so that the statement is true. Check each answer by expanding.

(a) $_____\,(3x - 4) = 12x^2 - 16x$

(b) $_____\,(2a + 5) = -14a - 35$

(c) $6x^2y\,(_____) = 18x^3y - 24x^2y^2$

(d) $-5m^2\,(_____) = -15m^4 + 20m^3 - 10m^2$

6. **The vanishing 1.** When a term is *exactly* the GCF, dividing it leaves 1 — and that 1 must be written. For $5x^2 + 5x$ the GCF is $5x$, and $5x \div 5x = 1$, so the factored form is $5x(x + 1)$, *not* $5x(x)$. Factor each polynomial; every one of them leaves a 1 or a -1 behind somewhere.
(a) $5x^2 + 5x$ (b) $7a^3 - 7a^2$ (c) $9y^3 + 3y^2 - 3y$ (d) $-6m^4 + 6m^3 - 6m^2$

7. **Error analysis.** Each student below made a different mistake. For each one, state what went wrong *and* write the correct factored form.
(a) Sam: $4x^2 + 6x = 2x(2x + 3x)$ (b) Kara: $5x^2 + 5x = 5x(x)$ (c) Dev: $12x^3 + 18x^2 = 2x^2(6x + 9)$

8. A polynomial has a GCF of $4x^2$, and one of its terms is $8x^3$.
(a) Write a possible three-term polynomial, then factor it. How do you know its GCF is exactly $4x^2$ and not something larger?
(b) Explain why a polynomial whose GCF is $4x^2$ can never contain the term $10x^5$.

9. Write a three-term polynomial whose GCF is $-3ab$, then factor it. Show the check.

Part C — When the Common Factor Is a Whole Bracket

A common factor does not have to be a monomial. In $3x(x+2) + 5(x+2)$ the two terms are $3x(x+2)$ and $5(x+2)$, and the bracket $(x + 2)$ sits inside both — so $(x + 2)$ *is* the GCF. Removing it works exactly as before: divide each term by $(x + 2)$ and collect what is left,

$$3x(x + 2) + 5(x + 2) \;=\; (x + 2)\big(3x + 5\big).$$

10. Factor each expression by removing the common binomial factor.
(a) $7x(x - 4) + 3(x - 4)$ (b) $2a(a + 5) - 9(a + 5)$ (c) $5m(m + 3) + (m + 3)$ (d) $4x(2x - 1) - 3(2x - 1)$

11. **Factoring by grouping.** A four-term polynomial often has no common factor at all, yet can still be factored. Split it into two pairs, remove the GCF from each pair, and if the same bracket appears twice, remove *it*:

$$x^3 + 2x^2 + 3x + 6 \;=\; x^2(x+2) + 3(x+2) \;=\; (x+2)\big(x^2+3\big).$$

Use grouping to factor each polynomial.
(a) $x^3 + 2x^2 + 3x + 6$ (b) $x^3 + 5x^2 + 4x + 20$ (c) $2x^3 + x^2 + 10x + 5$

12. Group and factor. Watch the signs: when the third term is negative, the GCF you remove from the second pair must be negative too.
(a) $x^3 - 3x^2 + 7x - 21$ (b) $6x^3 - 4x^2 - 15x + 10$

(c) $ab + 3a + 2b + 6$ (d) $4x^3 + 8x^2 - 3x - 6$

13. **Stretch: brackets that look different but are not.** The brackets $(x-3)$ and $(3-x)$ are opposites: $3-x = -(x-3)$. Factoring -1 out of one bracket turns it into the other, and a common factor appears. Factor each expression completely.
(a) $2x(x-3) - 5(3-x)$ (b) $4a(a-1) + 7(1-a)$ (c) $3y(2y-5) - 8(5-2y)$

Part D — Applications and Reasoning

14. Each expression below is the area of a rectangle. One side length is given; determine the other by removing that side as a factor.
(a) $A = 6x^2 + 9x$, width $3x$ (b) $A = 20x^3 - 15x^2$, width $5x^2$ (c) $A = 12x^2y + 18xy^2$, width $6xy$

15. A rectangular garden has area $A = 8x^2 + 20x$ square metres, where x is a positive number of metres.
(a) Factor A completely to give the dimensions of the garden. Why is $2x(4x + 10)$, although a true factoring, not the answer wanted?
(b) Find the two dimensions and the perimeter of the garden when $x = 5$, and check that your dimensions give the right area.

16. **Is it finished?** A factoring is **complete** when the expression left inside the bracket has no common factor other than 1. Decide whether each factoring below is complete. If it is not, finish the job.

(a) $3x(4x+8)$ (b) $5a^2(2a-3)$ (c) $2y(6y^2-9y)$ (d) $-4m(3m+5)$

17. Explain, in your own words, why the GCF of x^2 and x^5 is x^2 — the *lower* exponent — even though x^5 is the larger expression. Use the meaning of the word *factor* in your explanation.

18. Answer each.

(a) *(Multiple Choice)* Following the convention in this course, which is the complete factoring of $-20x^3+30x^2$?
A. $-10x^2(2x-3)$ B. $10x^2(-2x+3)$
C. $-5x^2(4x-6)$ D. $-10x(2x^2-3x)$

(b) *(Numerical Response)* When $24x^4-36x^3+60x^2$ is factored completely, the result has the form $ax^2\left(bx^2+cx+d\right)$. The value of $a+b+c+d$ is _____.

Answer Key. 1. (a) $12x^2y$ (b) $9a^2b$ (c) $-7m^2n$ (d) $15x$. **2.** (a) $6x\left(2x^3-3x^2+4x-1\right)$ (b) $10a^2\left(2a^3+3a^2-5a+1\right)$ (c) $9y\left(y^3-3y^2+2y-5\right)$. **3.** (a) $5a^2b^2(3a-5b+2)$ (b) $12x^2y^2(2x-3y+1)$ (c) $7m^2n^2\left(2m^2+3mn-4n^2\right)$. **4.** (a) $-4x\left(2x^2-3x+5\right)$ (b) $-6a^2\left(3a^2+4a-5\right)$ (c) $-6xy\left(2x^2-3xy+5y^2\right)$ (d) $-8m^2\left(2m^3-5m+3\right)$. **5.** (a) $4x$ (b) -7 (c) $3x-4y$ (d) $3m^2-4m+2$. **6.** (a) $5x(x+1)$ (b) $7a^2(a-1)$ (c) $3y\left(3y^2+y-1\right)$ (d) $-6m^2\left(m^2-m+1\right)$. **7.** (a) Sam divided $4x^2$ by $2x$ but not $6x$ — $6x\div 2x=3$, not $3x$; correctly $2x(2x+3)$. (b) Kara dropped the 1 left by $5x\div 5x$; correctly $5x(x+1)$. (c) Dev's factor is common but not *greatest* — $6x+9$ still has a factor of 3; correctly $6x^2(2x+3)$. **8.** (a) Answers vary; e.g. $16x^4+8x^3+12x^2=4x^2\left(4x^2+2x+3\right)$ — the coefficients 16, 8, 12 have GCF 4, the lowest power of x is x^2, and $4x^2+2x+3$ has no common factor left. (b) $10x^5\div 4x^2=\frac{5}{2}x^3$, which is not a monomial with a whole-number coefficient, so $4x^2$ is not a factor of $10x^5$. **9.** Answers vary; e.g. $-6a^2b+9ab^2-3ab=-3ab(2a-3b+1)$. Check: $-3ab(2a)=-6a^2b$, $-3ab(-3b)=9ab^2$, $-3ab(1)=-3ab$. **10.** (a) $(x-4)(7x+3)$ (b) $(a+5)(2a-9)$ (c) $(m+3)(5m+1)$ (d) $(2x-1)(4x-3)$. **11.** (a) $(x+2)\left(x^2+3\right)$ (b) $(x+5)\left(x^2+4\right)$ (c) $(2x+1)\left(x^2+5\right)$. **12.** (a) $x^2(x-3)+7(x-3)=(x-3)\left(x^2+7\right)$ (b) $2x^2(3x-2)-5(3x-2)=(3x-2)\left(2x^2-5\right)$ (c) $a(b+3)+2(b+3)=(b+3)(a+2)$ (d) $4x^2(x+2)-3(x+2)=(x+2)\left(4x^2-3\right)$. **13.** (a) $-5(3-x)=+5(x-3)$, so $2x(x-3)+5(x-3)=(x-3)(2x+5)$ (b) $7(1-a)=-7(a-1)$, so $4a(a-1)-7(a-1)=(a-1)(4a-7)$ (c) $-8(5-2y)=+8(2y-5)$, so $3y(2y-5)+8(2y-5)=(2y-5)(3y+8)$. **14.** (a) $6x^2+9x=3x(2x+3)$, length $2x+3$ (b) $20x^3-15x^2=5x^2(4x-3)$, length $4x-3$ (c) $12x^2y+18xy^2=6xy(2x+3y)$, length $2x+3y$. **15.** (a) $A=4x(2x+5)$, so the dimensions are $4x$ by $2x+5$. In $2x(4x+10)$ the bracket $4x+10$ still has a common factor of 2, so $2x$ is common but not *greatest* and the factoring is not complete. (b) At $x=5$: $4x=20$ m and $2x+5=15$ m (area 300 m^2, matching $8(25)+20(5)$); perimeter $=2(20+15)=70$ m. **16.** (a) Not complete — $4x+8=4(x+2)$; finished: $12x(x+2)$. (b) Complete. (c) Not complete — $6y^2-9y=3y(2y-3)$; finished: $6y^2(2y-3)$. (d) Complete. **17.** A factor must divide *both* expressions exactly. Since $x^2=x\cdot x$ and $x^5=x\cdot x\cdot x\cdot x\cdot x$, the copies of x they share number only two, so x^2 goes into both but x^5 does not go into x^2. The common part can never be larger than the smaller expression, so the GCF takes the lower exponent. **18.** (a) A. (B has a negative leading term inside the bracket, C is not the *greatest* common factor since $4x-6=2(2x-3)$, and D left an x behind in $2x^2-3x$.) (b) $24x^4-36x^3+60x^2=12x^2\left(2x^2-3x+5\right)$, so $a+b+c+d=12+2+(-3)+5=16$.

Lesson 3 — Extra Practice

Factoring Trinomials: $x^2 + bx + c$ with c Positive

Part A — Larger Constants: When the Search Is the Work

Every trinomial in this part has $c > 0$, so the two numbers share a sign — and that sign matches b. The only difficulty is that c now has many factor pairs. Work through the pairs in order ($1 \times c$, then $2 \times \ldots$, then $3 \times \ldots$) so you do not miss one.

1. Factor each trinomial. (Here b and c are both positive, so both numbers are positive.)
(a) $x^2 + 13x + 36$ (b) $x^2 + 17x + 72$ (c) $x^2 + 20x + 64$ (d) $x^2 + 15x + 56$

2. Factor each trinomial. (Here b is negative and c is positive, so **both** numbers must be negative.)
(a) $x^2 - 11x + 24$ (b) $x^2 - 20x + 64$ (c) $x^2 - 19x + 48$ (d) $x^2 - 22x + 72$

3. Factor each trinomial. In each of these the two numbers are very far apart, so the factor pair you want is near the start or the end of your list.
(a) $x^2 + 27x + 50$ (b) $x^2 - 26x + 48$ (c) $x^2 + 35x + 96$ (d) $x^2 - 29x + 100$

Part B — Common Factor First, Then the Trinomial

A trinomial is not factored completely until nothing more can come out. Remove the **GCF** first (Lesson 2), then factor whatever trinomial is left inside the brackets (Lesson 3). Both steps appear in the final answer.

4. Factor completely. Remove the common factor first, then factor the trinomial.
(a) $3x^2 + 21x + 30$ (b) $5x^2 - 45x + 90$ (c) $2x^2 + 26x + 84$

5. Factor completely. In part (c) the leading coefficient is negative, so remove a **negative** common factor.
(a) $4x^2 - 40x + 96$ (b) $7x^2 + 63x + 126$ (c) $-3x^2 - 21x - 36$

6. Factor completely. Here the common factor contains a variable.
(a) $x^3 + 12x^2 + 35x$ (b) $2x^3 - 24x^2 + 70x$ (c) $4ax^2 + 28ax + 48a$

Part C — Two Variables and Higher Powers

The sum–product rule does not care what the variable is called. In $x^2 + 7xy + 12y^2$ the pattern is still $(x + p\,y)(x + q\,y)$ with $pq = 12$ and $p + q = 7$; expand your answer to confirm the middle term comes back as $7xy$.

7. Factor each trinomial in two variables. Check one of them by expanding.
(a) $x^2 + 7xy + 12y^2$ (b) $a^2 - 13ab + 40b^2$ (c) $m^2 + 16mn + 63n^2$ (d) $p^2 - 15pq + 54q^2$

8. **Stretch.** In $x^4 + 7x^2 + 12$, notice that $x^4 = (x^2)^2$. If you let the "variable" be x^2, the expression is just $(\square)^2 + 7(\square) + 12$ and the usual rule applies. Factor each of the following into two brackets.
(a) $x^4 + 7x^2 + 12$ (b) $y^4 + 10y^2 + 21$ (c) $x^4 - 13x^2 + 36$

Part D — Knowing When to Stop

Not every trinomial factors over the integers. You may only say so after you have listed **every** factor pair of c and shown that none of them adds to b. A guess is not a reason.

9. Factor each trinomial if possible. If it cannot be factored over the integers, write "not factorable" and **list every factor pair of** c with its sum, as your justification.
(a) $x^2 + 5x + 7$ (b) $x^2 + 14x + 40$ (c) $x^2 - 7x + 11$

(d) $x^2 - 17x + 66$ (e) $x^2 + 8x + 14$ (f) $x^2 + 21x + 38$

10. Nadia says, "I tried 2×7 and 1×14 for $x^2 + 8x + 14$ and neither worked, so I will try some decimals next." Explain why decimals are not what is being asked for here, and state exactly how many factor pairs Nadia needed to check before she could safely stop.

Part E — Working Backwards

11. Consider $x^2 + bx + 15$, where b is an integer.

(a) List **every** integer value of b for which this trinomial can be factored over the integers.
(*Do not forget the negative values of b.*)

(b) Write the factored form that goes with each value of b you listed in part (a).

12. Consider $x^2 + 9x + c$, where c is a positive integer less than 20.

(a) List every value of c that makes the trinomial factorable over the integers, and give the factored form for each.

(b) The value $c = 20$ also works. Explain why $c = 20$ is the **largest** value of c that can ever work when $b = 9$ and both numbers are positive.

13. Complete each of the following.

(a) List every **positive** value of b for which $x^2 + bx + 40$ factors over the integers, then state the smallest and the largest.

(b) List every **positive** value of c for which $x^2 - 12x + c$ factors over the integers. One of your answers gives two identical brackets — which one?

Part F — Checking, Area, and Reasoning

14. **Error analysis.** Dev factors $x^2 + 10x + 21$ and gets $(x + 3)(x + 7)$. To check, he writes "$3 \times 7 = 21$, so the answer expands to $x^2 + 21$."

(a) Dev's factoring is correct but his check is not. Expand $(x+3)(x+7)$ properly with FOIL and show all four terms.

(b) Explain in one or two sentences which two terms Dev left out, and why a correct check must produce three terms, not two.

15. **Error analysis.** Priya factors $x^2 + 16x + 48$ and writes $(x + 6)(x + 8)$, saying "$6 \times 8 = 48$, which is the constant I need."

(a) Expand $(x + 6)(x + 8)$. Which of the two conditions — the product or the sum — did Priya satisfy, and which did she ignore?

(b) Give the correct factorization of $x^2 + 16x + 48$, and state the factor pair of 48 that she should have chosen.

16. A rectangle has an area of $x^2 + 12x + 35$ square metres.

(a) Factor the area to find binomial expressions for the length and the width.

(b) Write a simplified expression for the perimeter of the rectangle.

(c) If $x = 6$, find the length, the width, the area and the perimeter.

17. **Reasoning about signs.** Suppose $x^2 + bx + c$ factors as $(x + p)(x + q)$, where $c > 0$.

(a) Explain why p and q must have the **same** sign.

(b) If $b > 0$ as well, what sign are p and q? Justify your answer.

(c) If $b < 0$, what sign are p and q? Justify your answer.

18. Expand $(x + p)(x + q)$ using FOIL and collect like terms. Use your result to explain, in your own words, why the two numbers in the brackets must **multiply to** c and **add to** b.

Answer Key. 1. (a) $(x+4)(x+9)$ (b) $(x+8)(x+9)$ (c) $(x+4)(x+16)$ (d) $(x+7)(x+8)$. **2.** (a) $(x-3)(x-8)$ (b) $(x-4)(x-16)$ (c) $(x-3)(x-16)$ (d) $(x-4)(x-18)$. **3.** (a) $(x+2)(x+25)$ (b) $(x-2)(x-24)$ (c) $(x+3)(x+32)$ (d) $(x-4)(x-25)$. **4.** (a) $3(x+2)(x+5)$ (b) $5(x-3)(x-6)$ (c) $2(x+6)(x+7)$. **5.** (a) $4(x-4)(x-6)$ (b) $7(x+3)(x+6)$ (c) $-3(x+3)(x+4)$. **6.** (a) $x(x+5)(x+7)$ (b) $2x(x-5)(x-7)$ (c) $4a(x+3)(x+4)$. **7.** (a) $(x+3y)(x+4y)$ (b) $(a-5b)(a-8b)$ (c) $(m+7n)(m+9n)$ (d) $(p-6q)(p-9q)$. **8.** (a) $(x^2+3)(x^2+4)$ (b) $(y^2+3)(y^2+7)$ (c) $(x^2-4)(x^2-9)$ — each bracket here can be taken further once you meet the difference of squares in Lesson 5. **9.** (a) not factorable; pairs of 7: 1×7 (sum 8) only, and $8 \neq 5$ (b) $(x+4)(x+10)$ (c) not factorable; pairs of 11: -1×-11 (sum -12) only, and $-12 \neq -7$ (d) $(x-6)(x-11)$ (e) not factorable; pairs of 14: 1×14 (sum 15), 2×7 (sum 9), and neither is 8 (f) $(x+2)(x+19)$. **10.** The question asks for a factorization *over the integers*, so only whole-number factor pairs count; decimals would give infinitely many possibilities and would not be a factorization of the required kind. Since $14 = 1 \times 14 = 2 \times 7$, there are exactly two positive pairs, and with $b > 0$ and $c > 0$ both numbers must be positive — so after checking those two pairs she could stop and say "not factorable." **11.** (a) $b = 16,\ 8,\ -8,\ -16$. (b) $b = 16$: $(x+1)(x+15)$; $b = 8$: $(x+3)(x+5)$; $b = -8$: $(x-3)(x-5)$; $b = -16$: $(x-1)(x-15)$. **12.** (a) $c = 8$: $(x+1)(x+8)$; $c = 14$: $(x+2)(x+7)$; $c = 18$: $(x+3)(x+6)$. (b) The two positive numbers must add to 9, and of all such pairs the product is largest when the numbers are closest together — 4 and 5, giving $c = 20$. Any larger c has no positive pair summing to 9. **13.** (a) $b = 13,\ 14,\ 22,\ 41$ (from 5×8, 4×10, 2×20, 1×40); smallest 13, largest 41. (b) $c = 11,\ 20,\ 27,\ 32,\ 35,\ 36$; $c = 36$ gives $(x-6)(x-6) = (x-6)^2$. **14.** (a) $(x+3)(x+7) = x^2 + 7x + 3x + 21 = x^2 + 10x + 21$. (b) He kept only the First and Last products and dropped the two middle terms $7x$ and $3x$; those middle terms are exactly what build the bx term, so a correct check always gives three terms. **15.** (a) $(x+6)(x+8) = x^2 + 14x + 48$. She satisfied the *product* ($6 \times 8 = 48$) but ignored the *sum* ($6 + 8 = 14$, not 16). (b) $x^2 + 16x + 48 = (x+4)(x+12)$; the correct pair is 4 and 12, since $4 \times 12 = 48$ and $4 + 12 = 16$. **16.** (a) $x^2 + 12x + 35 = (x+5)(x+7)$, so the dimensions are $(x+5)$ m and $(x+7)$ m. (b) $P = 2\,[(x+5) + (x+7)] = 4x + 24$ m. (c) At $x = 6$: length 13 m, width 11 m, area 143 m^2, perimeter 48 m. **17.** (a) $pq = c$ and $c > 0$; a positive product comes only from two positives or two negatives, so p and q share a sign. (b) Both positive — they already share a sign, and two negatives would give $p + q = b < 0$. (c) Both negative — two positives would give $b > 0$, so the shared sign must be negative. **18.** $(x+p)(x+q) = x^2 + qx + px + pq = x^2 + (p+q)x + pq$. Matching this with $x^2 + bx + c$ term by term gives $b = p + q$ and $c = pq$: the constant term is the product of the two numbers, and the x-term is built by adding the two middle products, so the two numbers must add to b.

Lesson 4 — Extra Practice

Factoring Trinomials $x^2 + bx + c$, Part 2: c Negative

Part A — Bigger Constants, Same Rule

1. Factor each trinomial. The constants are larger, so list the factor pairs of c in order rather than guessing.
(a) $x^2 + 5x - 36$ (b) $x^2 - 7x - 60$ (c) $x^2 + 2x - 48$ (d) $x^2 - 13x - 48$

2. Factor each trinomial.
(a) $x^2 + x - 72$ (b) $x^2 - 9x - 70$ (c) $x^2 + 4x - 96$ (d) $x^2 - 6x - 91$

3. All three trinomials below use factor pairs of 80. Factor each one, then state which two parts use the *same* pair of numbers with the signs swapped.
(a) $x^2 + 16x - 80$ (b) $x^2 - 16x - 80$ (c) $x^2 - 2x - 80$

Part B — Deciding the Signs Before You Factor

4. For each trinomial, first state whether the two integers have the **same sign** or **opposite signs** and why. Then factor.
(a) $x^2 - 9x + 20$ (b) $x^2 - x - 20$ (c) $x^2 + 12x + 20$ (d) $x^2 + 8x - 20$

5. These are mixed: some have $c > 0$ and some have $c < 0$. Factor each one. One of them is a perfect square trinomial.
(a) $x^2 + 13x + 36$ (b) $x^2 - 5x - 36$ (c) $x^2 - 12x + 36$ (d) $x^2 + 9x - 36$

6. Factor completely. Remove the greatest common factor first, then factor what is left.
(a) $2x^2 - 4x - 30$ (b) $3x^2 + 3x - 90$ (c) $x^3 - 4x^2 - 45x$

7. **Two variables.** The same rule works when the last term carries a second variable: for $x^2 + bxy + cy^2$ you still need two integers with product c and sum b, and each one is attached to a y. Factor each expression, then expand to check.
(a) $x^2 - 3xy - 10y^2$ (b) $x^2 + 2xy - 35y^2$ (c) $a^2 - ab - 42b^2$

Part C — Prime Trinomials, Errors, and Working Backwards

8. Show that each trinomial below is **prime** over the integers. List every factor pair of the constant, give both sums each pair can produce, and write one sentence explaining why none of them works.

(a) $x^2 + 6x - 20$

(b) $x^2 - 3x - 15$

9. Factor where possible. If a trinomial is prime, write "prime."

(a) $x^2 + 7x - 44$

(b) $x^2 - 2x - 24$

(c) $x^2 + 4x - 22$

(d) $x^2 - 10x - 39$

10. **Error analysis.** Devon factors $x^2 - x - 12$ and writes $(x + 4)(x - 3)$, saying "4 times -3 is -12, so the product checks out." Expand his answer to show it is wrong, explain which condition he forgot to check, and give the correct factorization.

11. **Error analysis.** Jae factors $x^2+3x-18$ as $(x-6)(x+3)$, reasoning that 6 and 3 multiply to 18 and the signs must be opposite. Both of those statements are true, yet the answer is wrong. Explain the rule that decides *which* number gets the minus sign, then factor correctly.

12. **Work backwards.** Consider $x^2 + bx - 24$, where b is an integer.

(a) List every factor pair of 24.

(b) List every **positive** integer b that makes the trinomial factorable.

(c) List **all** integers b that work, and explain the pattern connecting your two lists.

13. Determine the missing number in each statement.

(a) $(x + 7)(x - ___) = x^2 + 2x - 35$

(b) $(x - ___)(x + 4) = x^2 - 5x - 36$

(c) $(x + 3)(x - ___) = x^2 - 9x - 36$

(d) $x^2 + ___x - 36 = (x + 12)(x - 3)$

14. Find every negative integer c greater than -20 for which $x^2 + 3x + c$ can be factored over the integers. Give the factorization in each case. *Hint:* if the two integers sum to 3, one of them is m and the other is $3 - m$.

Part D — Comparing, Applying, and Reasoning

15. Factor $x^2 + 4x - 21$ and $x^2 - 4x - 21$. Then describe **exactly** what changed between the two answers and what stayed the same.

16. A rectangle has an area of $x^2 + 3x - 40$ square metres, where x is measured in metres.

(a) Factor to find binomial expressions for the length and the width.

(b) Determine the length and width when $x = 12$, and verify the area.

(c) Explain why x must be greater than 5 for this to describe a real rectangle.

17. A rectangular prism has a volume of $x^3 + x^2 - 30x$ cubic centimetres.

(a) Factor completely to find three expressions for the dimensions.

(b) State the smallest integer value of x for which all three dimensions are positive.

18. **Reasoning.** Suppose $x^2 + bx + c$ factors as $(x + p)(x + q)$ and $c < 0$.

(a) Explain why p and q must have opposite signs. Refer to the product pq.

(b) Once you know the two numbers, explain how you decide which one gets the minus sign.

19. **Reasoning.** Can a trinomial with $c < 0$ ever be a perfect square trinomial — that is, factor as $(x + p)^2$ or $(x - p)^2$? Justify your answer using the sign of c.

20. **Stretch.** A trinomial such as $x^4 - 5x^2 - 36$ can be factored by treating x^2 as the variable: since $(x^2)^2 - 5(x^2) - 36$ needs two integers with product -36 and sum -5, it factors as $(x^2 - 9)(x^2 + 4)$. The first bracket is a difference of squares, so the complete factorization is $(x - 3)(x + 3)(x^2 + 4)$ — note that $x^2 + 4$ does not factor further over the integers. Use this method to factor each expression as completely as possible.

(a) $x^4 - 13x^2 + 36$

(b) $x^4 + 3x^2 - 18$

(c) $x^4 - 2x^2 - 24$

Answer Key. 1. (a) $(x+9)(x-4)$ (b) $(x-12)(x+5)$ (c) $(x+8)(x-6)$ (d) $(x-16)(x+3)$. **2.** (a) $(x+9)(x-8)$ (b) $(x-14)(x+5)$ (c) $(x+12)(x-8)$ (d) $(x-13)(x+7)$. **3.** (a) $(x+20)(x-4)$ (b) $(x-20)(x+4)$ (c) $(x-10)(x+8)$; parts (a) and (b) use the same pair 20 and 4 with the signs swapped. **4.** (a) same sign (product $+20$, sum -9, so both negative): $(x-5)(x-4)$ (b) opposite signs (product -20): $(x-5)(x+4)$ (c) same sign, both positive: $(x+2)(x+10)$ (d) opposite signs: $(x+10)(x-2)$. **5.** (a) $(x+4)(x+9)$ (b) $(x-9)(x+4)$ (c) $(x-6)^2$ — the perfect square (d) $(x+12)(x-3)$. **6.** (a) $2(x-5)(x+3)$ (b) $3(x+6)(x-5)$ (c) $x(x-9)(x+5)$. **7.** (a) $(x-5y)(x+2y)$ (b) $(x+7y)(x-5y)$ (c) $(a-7b)(a+6b)$. **8.** (a) Pairs of 20: $1, 20$; $2, 10$; $4, 5$, giving sums ± 19, ± 8, ± 1. None is 6, so it is prime. (b) Pairs of 15: $1, 15$; $3, 5$, giving sums ± 14, ± 2. None is -3, so it is prime. **9.** (a) $(x+11)(x-4)$ (b) $(x-6)(x+4)$ (c) prime (pairs of 22 give sums ± 21, ± 9) (d) $(x-13)(x+3)$. **10.** $(x+4)(x-3) = x^2+x-12$, not x^2-x-12. He checked only the product and not the sum: $4+(-3)=1$, but $b=-1$. The larger number must carry the sign of b, so the correct answer is $(x-4)(x+3)$. **11.** The number with the *larger absolute value* takes the same sign as b. Here $b=+3$, so the 6 must be positive and the 3 negative: $x^2+3x-18=(x+6)(x-3)$. Jae's answer expands to $x^2-3x-18$. **12.** (a) $1, 24$; $2, 12$; $3, 8$; $4, 6$. (b) $b = 2,\ 5,\ 10,\ 23$. (c) All integers: $b = \pm 2,\ \pm 5,\ \pm 10,\ \pm 23$. Each factor pair gives one positive and one negative b (the difference of the pair, with either sign), since swapping which number is negative reverses the sum. **13.** (a) 5 (b) 9 (c) 12 (d) 9. **14.** $c=-4$: $(x-1)(x+4)$; $c=-10$: $(x-2)(x+5)$; $c=-18$: $(x-3)(x+6)$. (Taking $m=-1,-2,-3$ gives $c=m(3-m)=-4,-10,-18$; $m=-4$ would give $c=-28$, which is below -20.) **15.** $x^2+4x-21=(x+7)(x-3)$ and $x^2-4x-21=(x-7)(x+3)$. The two numbers are still 7 and 3 and the product is still -21; only the signs swapped, because b changed sign and the larger number (7) always follows the sign of b. **16.** (a) $(x+8)(x-5)$, so the dimensions are $x+8$ and $x-5$ metres. (b) Length 20 m, width 7 m; $20\times 7=140$, and $12^2+3(12)-40=140$ m^2. ✓ (c) If $x\le 5$ the width $x-5$ is zero or negative, which is impossible for a real side length; only $x>5$ makes both dimensions positive. **17.** (a) $x(x+6)(x-5)$, so the dimensions are x, $x+6$ and $x-5$ cm. (b) $x=6$ (at $x=5$ one dimension would be 0). **18.** (a) $pq=c$, and a product of two integers is negative only when one factor is positive and the other is negative — two positives or two negatives both give a positive product. (b) The number with the larger absolute value gets the sign of b, because it dominates the sum $p+q=b$; the smaller number takes the opposite sign. **19.** No. A perfect square expands to $x^2\pm 2px+p^2$, and $p^2\ge 0$, so c can never be negative. (Equivalently, a square needs the two numbers to be equal, but with $c<0$ they must have opposite signs.) **20.** (a) $(x^2-4)(x^2-9)=(x-2)(x+2)(x-3)(x+3)$ (b) $(x^2+6)(x^2-3)$ — neither bracket factors further over the integers (c) $(x^2-6)(x^2+4)$, and neither bracket factors further over the integers.

Lesson 5 — Extra Practice

Factoring the Difference of Squares

Part A — Leading Coefficients and Two Variables

1. Factor each difference of squares. Take the square root of *both* the coefficient and the variable part.
(a) $49x^2 - 64$ (b) $100a^2 - 9b^2$ (c) $25m^2 - 121n^2$ (d) $144p^2 - 169q^2$

2. Factor each difference of squares. Watch for the ones written with the constant first.
(a) $81 - 64y^2$ (b) $36x^2 - 169$ (c) $4a^2 - 225b^2$ (d) $1 - 196t^2$

3. Factor each difference of squares. Every exponent here is even, so every term is a perfect square.
(a) $9x^4 - 49$ (b) $64m^6 - 25n^2$ (c) $121a^2b^2 - 16c^4$

Part B — Higher Powers: Factoring Twice

Some binomials are a difference of squares *twice*. Consider $x^4 - 16$. Since $x^4 = (x^2)^2$ and $16 = 4^2$, the first pass gives $x^4 - 16 = (x^2 - 4)(x^2 + 4)$. Now look at each bracket: $x^2 - 4$ is itself a difference of squares and factors again, while $x^2 + 4$ is a *sum* of squares and stops. The complete factorization is $x^4 - 16 = (x - 2)(x + 2)(x^2 + 4)$. **Always re-check each factor you produce.**

4. Work through $x^4 - 81$ one stage at a time.
(a) Write x^4 and 81 each as a square, then factor once.
(b) Which of your two brackets can be factored again? Which one cannot, and why?
(c) Write the complete factorization of $x^4 - 81$.

5. Factor completely. Each of these needs more than one pass.
(a) $16x^4 - 81$ (b) $x^4 - 1$

(c) $a^4 - b^4$ (d) $81x^4 - 16y^4$

6. Factor completely. Decide for each factor you write whether it can go further.
(a) $m^4 - 625$ (b) $256 - x^4$

Part C — Common Factor First

7. None of these is a difference of squares as written. Remove the greatest common factor first, then factor completely.
(a) $18x^2 - 50$ (b) $x^3 - 25x$ (c) $12a^3 - 75a$ (d) $98m^2 - 8n^2$

8. Factor completely. These need a GCF *and* two passes of the difference of squares.
(a) $2x^4 - 32$ (b) $3x^5 - 3x$ (c) $5x^4 - 80y^4$

Part D — Is It a Difference of Squares?

9. Decide whether each expression can be factored using the difference of squares. If it can, factor it completely. If it cannot, **give the reason** — name which of the two conditions (subtraction, and both terms perfect squares) fails.
(a) $x^2 + 16$ (b) $x^2 - 7$ (c) $x^3 - 8$

(d) $4x^2 - 9$ (e) $9x^2 + 25y^2$ (f) $50x^2 - 18$

10. A student writes $x^2 - 9 = (x - 3)^2$. Expand $(x - 3)^2$ to show this is wrong, then give the correct factorization.

11. Another student claims $x^2 + 25 = (x + 5)(x - 5)$. Expand the right side, state what it actually equals, and explain what $x^2 + 25$ factors to instead.

12. Devin factors $4x^4 - 64$ like this: $4x^4 - 64 = 4(x^4 - 16) = 4(x^2 - 4)(x^2 + 4)$, and stops. Has he factored **completely**? If not, explain what he missed and finish the job.

Part E — Working Backwards and Mental Arithmetic

13. Determine the missing expression in each equation.
(a) $25x^2 - 49 = (5x - 7)(_____)$ (b) $_____ - 81 = (y - 9)(y + 9)$ (c) $16a^2 - _____ = (4a - 11)(4a + 11)$

14. In the equation $\square^2 - 49 = (\square - 7)(\square + 7)$, the same expression fills every box.
(a) What goes in the boxes if the left side is $36x^2 - 49$?
(b) What goes in the boxes if the left side is $x^4 - 49$?
(c) What goes in the boxes if the left side is $81a^2b^2 - 49$?

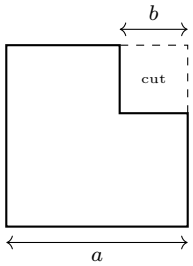
15. Consider the binomial $9x^2 - c$, where c is a natural number.
(a) List every value of c from 1 to 100 that makes $9x^2 - c$ a difference of squares with integer factors.
(b) Explain why $c = 10$ is not on your list.
(c) Factor the case $c = 64$.

16. The pattern also works on plain numbers: $83^2 - 17^2 = (83 - 17)(83 + 17) = 66 \times 100 = 6600$, which is far faster than squaring both numbers. Evaluate each of the following **without a calculator**, showing the factored form first.
(a) $75^2 - 25^2$ (b) $101^2 - 99^2$ (c) $250^2 - 150^2$ (d) $2024^2 - 2023^2$

17. Use the pattern to show that the difference between the squares of any two consecutive whole numbers, $n^2 - (n - 1)^2$, is always the odd number $2n - 1$. Then use your result to state $501^2 - 500^2$ instantly.

Part F — Geometry, Reasoning, and Stretch

18. A square sheet of metal has side length a. A smaller square of side length b is cut from one corner and discarded.



(a) Write an expression for the area of the remaining metal, then factor it.

(b) Cut the remaining L-shape into two rectangles. Give the dimensions of each, and show their areas add to $(a-b)(a+b)$.

(c) Check your formula when $a = 17$ cm and $b = 3$ cm by computing both forms.

19. Explain why a **sum** of squares such as $x^2 + 36$ can never be factored into two binomials with integer coefficients. *Hint:* if $x^2 + 36 = (x+p)(x+q)$, then expanding gives $p + q = 0$ and $pq = 36$. Show that no pair of integers can do both.

20. **Stretch.** A whole bracket can play the role of a in $a^2 - b^2$. For example, $(x+1)^2 - 9$ has $a = (x+1)$ and $b = 3$, so it factors as $\big((x+1)-3\big)\big((x+1)+3\big)$. Factor each of the following completely and simplify each bracket.

(a) $(x+1)^2 - 9$ (b) $(2y-5)^2 - 16$ (c) $(x+3)^2 - (x-3)^2$

21. **Stretch.** Factor $x^8 - 1$ completely over the integers. How many passes of the difference of squares does it take, and what stops the process?

Answer Key. 1. (a) $(7x-8)(7x+8)$ (b) $(10a-3b)(10a+3b)$ (c) $(5m-11n)(5m+11n)$ (d) $(12p-13q)(12p+13q)$. **2.** (a) $(9-8y)(9+8y)$ (b) $(6x-13)(6x+13)$ (c) $(2a-15b)(2a+15b)$ (d) $(1-14t)(1+14t)$. **3.** (a) $(3x^2-7)(3x^2+7)$ (b) $(8m^3-5n)(8m^3+5n)$ (c) $(11ab-4c^2)(11ab+4c^2)$. **4.** (a) $x^4-81 = (x^2)^2-9^2 = (x^2-9)(x^2+9)$ (b) x^2-9 factors again; x^2+9 is a sum of squares and does not factor over the integers (c) $(x-3)(x+3)(x^2+9)$. **5.** (a) $(2x-3)(2x+3)(4x^2+9)$ (b) $(x-1)(x+1)(x^2+1)$ (c) $(a-b)(a+b)(a^2+b^2)$ (d) $(3x-2y)(3x+2y)(9x^2+4y^2)$. **6.** (a) $(m-5)(m+5)(m^2+25)$ (b) $(4-x)(4+x)(16+x^2)$. **7.** (a) $2(3x-5)(3x+5)$ (b) $x(x-5)(x+5)$ (c) $3a(2a-5)(2a+5)$ (d) $2(7m-2n)(7m+2n)$. **8.** (a) $2(x-2)(x+2)(x^2+4)$ (b) $3x(x-1)(x+1)(x^2+1)$ (c) $5(x-2y)(x+2y)(x^2+4y^2)$. **9.** (a) no — it is a *sum* of squares, not a difference (b) no — it is a difference, but 7 is not a perfect square (c) no — x^3 and 8 are cubes, not squares (an odd exponent is never a perfect square) (d) yes, $(2x-3)(2x+3)$ (e) no — sum of squares (f) yes, after the GCF: $2(25x^2-9) = 2(5x-3)(5x+3)$. **10.** $(x-3)^2 = (x-3)(x-3) = x^2-6x+9$, which is not x^2-9. The two factors must have *opposite* signs: $x^2-9 = (x-3)(x+3)$. **11.** $(x+5)(x-5) = x^2-25$, not x^2+25. A sum of squares such as x^2+25 cannot be factored over the integers, so it is already in simplest form. **12.** No. He stopped with x^2-4 still factorable. Completely: $4x^4-64 = 4(x-2)(x+2)(x^2+4)$. **13.** (a) $(5x+7)$ (b) y^2 (c) 121. **14.** (a) $6x$ (b) x^2 (c) $9ab$. **15.** (a) c must be a perfect square: $1, 4, 9, 16, 25, 36, 49, 64, 81, 100$ (b) 10 is not a perfect square, so $\sqrt{10}$ is not an integer and the factors would not have integer terms (c) $9x^2-64 = (3x-8)(3x+8)$. **16.** (a) $(75-25)(75+25) = 50 \times 100 = 5000$ (b) $(101-99)(101+99) = 2 \times 200 = 400$ (c) $(250-150)(250+150) = 100 \times 400 = 40\,000$ (d) $(2024-2023)(2024+2023) = 1 \times 4047 = 4047$. **17.** $n^2-(n-1)^2 = \big(n-(n-1)\big)\big(n+(n-1)\big) = (1)(2n-1) = 2n-1$. With $n = 501$: $501^2-500^2 = 2(501)-1 = 1001$. **18.** (a) $a^2-b^2 = (a-b)(a+b)$ (b) one cut gives an a by $(a-b)$ rectangle and a b by $(a-b)$ rectangle; $a(a-b)+b(a-b) = (a-b)(a+b)$ (c) $17^2-3^2 = 289-9 = 280$ and $(17-3)(17+3) = 14 \times 20 = 280$ — they agree. **19.** If $x^2+36 = (x+p)(x+q) = x^2+(p+q)x+pq$, then $p+q = 0$ and $pq = 36$. From $p+q = 0$, $q = -p$, so $pq = -p^2$, which is never positive. So $pq = 36$ is impossible and no such integers exist. **20.** (a) $(x-2)(x+4)$ (b) $(2y-9)(2y-1)$ (c) $(6)(2x) = 12x$. **21.** $x^8-1 = (x^4-1)(x^4+1) = (x^2-1)(x^2+1)(x^4+1) = (x-1)(x+1)(x^2+1)(x^4+1)$ — three passes. The process stops because x^2+1 and x^4+1 are sums of squares.

Lesson 6 — Extra Practice

Factoring Perfect Square Trinomials

Part A — Leading Coefficients and Two Variables

1. Factor each perfect square trinomial. Take the square root of the *whole* first term, coefficient included.
(a) $9x^2 + 30x + 25$ (b) $49x^2 - 84x + 36$ (c) $16x^2 + 40x + 25$ (d) $25x^2 - 20x + 4$

2. Factor each perfect square trinomial. Check your answer by confirming that the middle term equals $2 \times \sqrt{\text{first}} \times \sqrt{\text{last}}$.
(a) $4x^2 + 36x + 81$ (b) $64x^2 - 16x + 1$ (c) $36x^2 + 84x + 49$ (d) $81x^2 - 90x + 25$

3. Each trinomial below has **two** variables. The pattern is unchanged: the first and last terms are squares, and the middle term is twice the product of their roots.
(a) $4a^2 - 12ab + 9b^2$ (b) $25m^2 + 60mn + 36n^2$ (c) $49p^2 - 28pq + 4q^2$

Part B — The Recognition Test

Two perfect squares at the ends are *not* enough. A trinomial $ax^2 + bx + c$ is a perfect square only when the middle term is exactly $2\sqrt{a}\sqrt{c}$ — for $16x^2 + 20x + 25$ the ends are fine ($4x$ and 5), but $2(4x)(5) = 40x$, not $20x$, so it fails. In each question below, decide first, then act.

4. State whether each trinomial is a perfect square trinomial. If it is, factor it. If it is not, state the middle term it *would* need in order to be one.
(a) $9x^2 + 12x + 4$ (b) $16x^2 + 20x + 25$ (c) $25x^2 - 30x + 9$ (d) $4x^2 + 10x + 25$

5. State whether each is a perfect square trinomial and say **which part of the test** fails when it is not. Then factor completely if the expression factors at all.
(a) $x^2 + 13x + 36$ (b) $36x^2 - 12x + 1$ (c) $4x^2 + 16x + 12$ (d) $25a^2 + 20ab + 16b^2$

6. Dmitri factored $x^2+10x+25$ as $(x+5)(x-5)$. Multiply his answer out to show it is wrong, explain what pattern he confused it with, and give the correct factorization.

7. Hana says $x^2 + 13x + 36$ must be a perfect square trinomial "because 36 is a perfect square." Explain why her reasoning fails the three-part test, then factor the trinomial correctly.

Part C — Common Factor First

A trinomial can hide a perfect square behind a common factor. Always remove the greatest common factor before you run the test: $2x^2 + 12x + 18 = 2(x^2 + 6x + 9) = 2(x + 3)^2$.

8. Factor completely. Remove the greatest common factor first, then factor what is left.

(a) $2x^2 + 12x + 18$ (b) $3x^3 - 24x^2 + 48x$ (c) $5x^2 - 70x + 245$ (d) $4x^2 + 24x + 36$

9. Factor completely. In these the leftover trinomial still carries a leading coefficient.

(a) $12x^3 + 60x^2 + 75x$ (b) $2a^2 - 20ab + 50b^2$ (c) $50x^2 + 40x + 8$

Part D — Filling in the Missing Piece

10. Replace each $\square$ so the result is a perfect square trinomial, then write the factored form. Part (c) has **two** correct answers — give both.

(a) $x^2 + 14x + \square$ (b) $\square x^2 - 20x + 25$ (c) $9x^2 + \square + 49$

11. Replace each $\square$ so the result is a perfect square trinomial. Part (a) again has two answers.

(a) $25x^2 + \square + 9$ (b) $16x^2 - 56x + \square$ (c) $x^2 - \square + 121$ (d) $\square + 12ab + 9b^2$

12. *(Numerical Response)* The trinomial $25x^2 + Kx + 64$ is a perfect square trinomial, where K is positive. The value of K is _____.

13. *(Multiple Choice)* Which trinomial is **not** a perfect square trinomial?

A. $49x^2 + 14x + 1$ B. $4x^2 + 12x + 9$
C. $9x^2 + 6x + 4$ D. $x^2 - 6x + 9$

Part E — Structure, Geometry, and Reasoning

14. The same two numbers, x and 4, build two very different expressions.

(a) Factor $x^2 - 16$ and factor $x^2 - 8x + 16$.

(b) Explain in your own words why one is a difference of squares and the other is a perfect square trinomial, and what the presence of the middle term tells you about which pattern to use.

15. A square has area $(x^2 + 16x + 64)$ cm^2. Determine an expression for its side length and an expression for its perimeter. Then find both numerically when $x = 4$.

16. A square tile has area $(4x^2 + 20x + 25)$ mm^2. Determine an expression for its side length and an expression for its perimeter. Then find both numerically when $x = 3$.

17. **Reasoning.** Answer each part in a full sentence, using the three-part test to justify your claim.

(a) Why must the constant term of a perfect square trinomial always be positive?

(b) Can the middle term of a perfect square trinomial be negative? What does that change in the factored form?

(c) Can the *first* term be negative? What would you do with $-x^2+6x-9$?

18. **Stretch.** Treat the bracket as a single object. If you let $u = (x+1)$, then $(x+1)^2+6(x+1)+9$ becomes $u^2+6u+9 = (u+3)^2$. Substituting back gives $(x + 1 + 3)^2 = (x + 4)^2$. Use that idea below, and expand each answer to check.

(a) Factor $(x+1)^2+6(x+1)+9$, then expand the original expression to confirm your answer.

(b) Factor $(x + 3)^2 - 10(x + 3) + 25$, then expand the original expression to confirm your answer.

Answer Key. 1. (a) $(3x + 5)^2$ (b) $(7x - 6)^2$ (c) $(4x + 5)^2$ (d) $(5x - 2)^2$. **2.** (a) $(2x + 9)^2$ (b) $(8x - 1)^2$ (c) $(6x + 7)^2$ (d) $(9x - 5)^2$. **3.** (a) $(2a - 3b)^2$ (b) $(5m + 6n)^2$ (c) $(7p - 2q)^2$. **4.** (a) yes, $(3x + 2)^2$ (b) no; it would need $40x$ (c) yes, $(5x - 3)^2$ (d) no; it would need $20x$. **5.** (a) not a PST — the middle term would have to be $12x$, not $13x$; it still factors as $(x + 4)(x + 9)$ (b) PST, $(6x - 1)^2$ (c) not a PST — 12 is not a perfect square; common factor first: $4(x + 1)(x + 3)$ (d) not a PST — the ends are $5a$ and $4b$, so the middle would have to be $40ab$; it does not factor. **6.** $(x+5)(x-5) = x^2-25$, not $x^2+10x+25$; he used the difference-of-squares pattern, which has no middle term. Correctly, $x^2+10x+25 = (x+5)^2$. **7.** A perfect square last term is only one of three conditions — the middle term must also equal $2(x)(6) = 12x$, and $13x \neq 12x$, so the test fails. Factoring by sum and product, $x^2 + 13x + 36 = (x + 4)(x + 9)$. **8.** (a) $2(x + 3)^2$ (b) $3x(x - 4)^2$ (c) $5(x - 7)^2$ (d) $4(x + 3)^2$. **9.** (a) $3x(2x + 5)^2$ (b) $2(a - 5b)^2$ (c) $2(5x + 2)^2$. **10.** (a) 49; $(x + 7)^2$ (b) 4; $(2x - 5)^2$ (c) $+42x$ giving $(3x + 7)^2$, or $-42x$ giving $(3x - 7)^2$. **11.** (a) $+30x$ giving $(5x + 3)^2$, or $-30x$ giving $(5x - 3)^2$ (b) 49; $(4x - 7)^2$ (c) $22x$; $(x - 11)^2$ (d) $4a^2$; $(2a + 3b)^2$. **12.** 80. **13.** C (the middle term would have to be $12x$). **14.** (a) $x^2 - 16 = (x - 4)(x + 4)$ and $x^2 - 8x + 16 = (x - 4)^2$ (b) $x^2 - 16$ has two terms and a subtraction between two squares, so the factors are a sum and a difference; $x^2 - 8x + 16$ has a middle term of $2(x)(4) = 8x$, which is exactly the sign that the same binomial is used twice. If a middle term is present, test for a perfect square; if it is absent, look for a difference of squares. **15.** Side $= (x + 8)$ cm, perimeter $= (4x + 32)$ cm; at $x = 4$, side = 12 cm, perimeter = 48 cm (area 144 cm^2). **16.** Side $= (2x + 5)$ mm, perimeter $= (8x + 20)$ mm; at $x = 3$, side = 11 mm, perimeter = 44 mm (area 121 mm^2). **17.** (a) The constant is the square of the second root, and squaring any real number gives a result that is never negative — so it is always positive (never subtracted). (b) Yes. A negative middle term means the binomial being squared is a *difference*, as in $x^2 - 8x + 16 = (x - 4)^2$; only the sign inside the bracket changes, and the constant stays positive. (c) Not in the standard form — but a negative leading term just means -1 is a common factor: $-x^2 + 6x - 9 = -(x^2 - 6x + 9) = -(x - 3)^2$. **18.** (a) $(x + 1 + 3)^2 = (x + 4)^2$; expanding the original gives $x^2 + 2x + 1 + 6x + 6 + 9 = x^2 + 8x + 16 = (x + 4)^2$ ✓ (b) $(x + 3 - 5)^2 = (x - 2)^2$; expanding the original gives $x^2 + 6x + 9 - 10x - 30 + 25 = x^2 - 4x + 4 = (x - 2)^2$ ✓

Lesson 7 — Extra Practice

Factoring $ax^2 + bx + c$ (where $a \neq 1$)

Part A — The Decomposition Search with Larger Numbers

1. Factor $8x^2 + 26x + 15$ by decomposition. Answer each step in order — this is the whole method, written out once.
(a) Write the value of $a \times c$.
(b) List every pair of positive integers whose product is that value, then circle the pair whose sum is $b = 26$.
(c) Use the circled pair to split $26x$ and write the four-term expression.
(d) Group the four terms into two pairs and factor the GCF out of each pair.
(e) Factor out the common binomial, then check by expanding.

2. Factor by decomposition. With ac this large, list the factor pairs of ac in order so you do not miss one.
(a) $6x^2 + 19x + 10$ (b) $12x^2 + 25x + 12$

3. Factor each trinomial by decomposition.
(a) $10x^2 + 29x + 21$ (b) $9x^2 + 21x + 10$ (c) $15x^2 + 22x + 8$

Part B — Negative Coefficients and a Negative Leading Term

4. Factor by decomposition. Watch the signs when you choose the two integers.
(a) $12x^2 - 25x + 12$ (b) $8x^2 + 2x - 15$

5. Factor each trinomial by decomposition.
(a) $6x^2 - 17x + 5$ (b) $10x^2 - x - 21$ (c) $9x^2 - 9x - 4$

6. When a is negative, factor out -1 (together with any other common factor) *first*, then decompose the trinomial that is left. Leave the -1 in front of your final answer.
(a) $-2x^2 + 5x + 12$ (b) $-6x^2 - x + 15$ (c) $-4x^2 + 16x - 15$

Part C — GCF First, Two Variables, and Higher Powers

7. Factor completely. Remove the GCF before you decompose — it makes $a \times c$ much smaller.
(a) $6x^2 + 21x + 9$ (b) $12x^3 - 2x^2 - 4x$

8. Factor completely.
(a) $20x^2 - 65x + 50$ (b) $18x^3 + 33x^2 - 30x$ (c) $50x^2 - 40x + 8$

9. These trinomials contain two variables. Decomposition works exactly the same way: use $a \times c$ where c is the coefficient of y^2, and carry a y along with the second pair of terms.
(a) $6x^2 + 7xy - 3y^2$ (b) $10x^2 - 11xy + 3y^2$ (c) $8x^2 + 10xy - 25y^2$

10. **Stretch.** Each expression below is a trinomial in x^2 rather than in x. Decompose it exactly as usual, treating x^2 the way you normally treat x.
(a) $6x^4 + 7x^2 - 3$ (b) $4x^4 - 8x^2 - 5$

Part D — Not Factorable, Error Analysis, and Working Backwards

11. Two of these trinomials factor over the integers and two do not. Factor the ones that do. For the ones that do not, list *every* integer pair whose product is $a \times c$ and show that no pair adds to b.
(a) $4x^2 + 5x + 3$ (b) $6x^2 + 11x + 4$ (c) $9x^2 - 6x + 5$ (d) $8x^2 - 14x + 3$

12. **Error analysis.** Jordan factored $6x^2 - 7x - 20$. He found $ac = -120$ and the correct pair 8 and -15, then wrote:

$$6x^2 - 7x - 20 = 6x^2 + 8x - 15x - 20 = 2x(3x + 4) - 5(3x - 4),$$

and concluded that the expression cannot be factored because the two binomials are different.

(a) Identify the exact step where Jordan went wrong, and explain what he did to the signs.

(b) Redo the grouping correctly and give the factored form. Check it by expanding.

13. **Error analysis.** Sam skipped the GCF step and factored $4x^2 + 14x + 6$ straight into $(2x + 6)(2x + 1)$. When she expanded to check, she got $4x^2 + 14x + 6$ back, so she stopped.

(a) Her check really does work. Explain why her answer is still not acceptable as a complete factorization.

(b) Finish the job and write $4x^2 + 14x + 6$ in fully factored form. Then redo it the intended way — GCF first — and confirm you get the same answer.

14. **Work backwards.** Find the missing binomial in each statement. You can divide, or you can reason from the first and last terms and then check the middle term.

(a) $(2x + 3)(\square) = 6x^2 + x - 12$ (b) $(3x - 2)(\square) = 15x^2 + 11x - 14$ (c) $(\square)(2x - 7) = 10x^2 - 29x - 21$

15. **Work backwards.** For each trinomial, find **every** integer value of b that makes it factorable over the integers. *Hint:* list the factor pairs of $a \times c$; each pair's sum is a possible b, and each pair also works with both signs reversed.

(a) $4x^2 + bx + 9$

(b) $6x^2 + bx + 5$

Part E — Applications and a Closer Look

16. A rectangular patio has an area of $10x^2 + 11x - 6$ square metres, and both its length and its width can be written as binomials with integer coefficients.

(a) Factor the area to find expressions for the length and the width.

(b) Determine the length and the width when $x = 6$, and verify that they multiply to the area.

(c) Determine the perimeter of the patio when $x = 6$.

17. **Two ways to see the same thing.**

(a) Factor $4x^2 - 12x + 9$ using decomposition, showing ac, the integer pair, the four-term form, and the grouping.

(b) Look at your answer. Why can it be written as a square of a single binomial?

(c) A trinomial $ax^2 + bx + c$ is a perfect square exactly when a and c are perfect squares and $b = \pm 2\sqrt{a}\sqrt{c}$. Use that test to decide quickly whether $9x^2 + 30x + 25$ and $16x^2 - 24x + 9$ are perfect squares, then write each in squared form.

Answer Key. 1. (a) $ac = 120$ (b) $1 \cdot 120$, $2 \cdot 60$, $3 \cdot 40$, $4 \cdot 30$, $5 \cdot 24$, $6 \cdot 20$, $8 \cdot 15$, $10 \cdot 12$; the pair is 6 and 20 (c) $8x^2 + 6x + 20x + 15$ (d) $2x(4x + 3) + 5(4x + 3)$ (e) $(4x + 3)(2x + 5)$; check $8x^2 + 10x + 12x + 15 = 8x^2 + 26x + 15$. **2.** (a) $ac = 60$, pair 4 and 15: $(2x + 5)(3x + 2)$ (b) $ac = 144$, pair 9 and 16: $(3x + 4)(4x + 3)$. **3.** (a) $(2x + 3)(5x + 7)$ (b) $(3x + 2)(3x + 5)$ (c) $(3x + 2)(5x + 4)$. **4.** (a) $ac = 144$, pair -9 and -16: $(3x - 4)(4x - 3)$ (b) $ac = -120$, pair 12 and -10: $(2x + 3)(4x - 5)$. **5.** (a) $(2x - 5)(3x - 1)$ (b) $(2x - 3)(5x + 7)$ (c) $(3x - 4)(3x + 1)$. **6.** (a) $-(2x^2 - 5x - 12) = -(x - 4)(2x + 3)$ (b) $-(6x^2 + x - 15) = -(2x - 3)(3x + 5)$ (c) $-(4x^2 - 16x + 15) = -(2x - 3)(2x - 5)$. **7.** (a) $3(2x^2 + 7x + 3) = 3(x + 3)(2x + 1)$ (b) $2x(6x^2 - x - 2) = 2x(2x + 1)(3x - 2)$. **8.** (a) $5(x - 2)(4x - 5)$ (b) $3x(2x + 5)(3x - 2)$ (c) $2(5x - 2)^2$. **9.** (a) $(2x + 3y)(3x - y)$ (b) $(2x - y)(5x - 3y)$ (c) $(2x + 5y)(4x - 5y)$. **10.** (a) $ac = -18$, pair 9 and -2: $(2x^2 + 3)(3x^2 - 1)$ (b) $ac = -20$, pair -10 and 2: $(2x^2 - 5)(2x^2 + 1)$. **11.** (a) *Not factorable.* $ac = 12$ and $b = 5$; the positive pairs are $1, 12$ (sum 13), $2, 6$ (sum 8), $3, 4$ (sum 7), and negative pairs give negative sums — none add to 5. (b) $(2x + 1)(3x + 4)$ (c) *Not factorable.* $ac = 45$ and $b = -6$; the negative pairs are $-1, -45$ (sum -46), $-3, -15$ (sum -18), $-5, -9$ (sum -14), and positive pairs give positive sums — none add to -6. (d) $(2x - 3)(4x - 1)$. **12.** (a) The error is in the second group. $-15x - 20$ has GCF -5, and $-15x \div (-5) = 3x$ while $-20 \div (-5) = 4$, so the bracket must be $(3x + 4)$, not $(3x - 4)$ — he divided -20 by -5 and wrote -4. (b) $2x(3x + 4) - 5(3x + 4) = (3x + 4)(2x - 5)$; check $6x^2 - 15x + 8x - 20 = 6x^2 - 7x - 20$. ✓ **13.** (a) The factor $2x + 6$ still has a common factor of 2, so the expression is not factored *completely*; expanding correctly only shows the product is right, not that it is finished. (b) $(2x + 6)(2x + 1) = 2(x + 3)(2x + 1)$. GCF first: $4x^2 + 14x + 6 = 2(2x^2 + 7x + 3) = 2(x + 3)(2x + 1)$ — the same answer. **14.** (a) $3x - 4$ (b) $5x + 7$ (c) $5x + 3$. **15.** (a) $ac = 36$; pairs $1, 36$; $2, 18$; $3, 12$; $4, 9$; $6, 6$ give sums 37, 20, 15, 13, 12, and the negatives of those pairs give the opposites, so $b = \pm 12$, ± 13, ± 15, ± 20, ± 37 (ten values). (b) $ac = 30$; pairs $1, 30$; $2, 15$; $3, 10$; $5, 6$ give sums 31, 17, 13, 11, so $b = \pm 11$, ± 13, ± 17, ± 31 (eight values). **16.** (a) $10x^2 + 11x - 6 = (2x + 3)(5x - 2)$, so the dimensions are $(5x - 2)$ m by $(2x + 3)$ m. (b) $x = 6$: length 28 m, width 15 m; $28 \times 15 = 420$ m^2, and $10(36) + 11(6) - 6 = 360 + 66 - 6 = 420$ m^2. ✓ (c) $P = 2(28 + 15) = 86$ m. **17.** (a) $ac = 36$, pair -6 and -6: $4x^2 - 6x - 6x + 9 = 2x(2x - 3) - 3(2x - 3) = (2x - 3)(2x - 3)$. (b) Both factors are identical, so the product is $(2x - 3)^2$ — this happens whenever the two integers in the decomposition are equal. (c) $9x^2 + 30x + 25$: $a = 9$ and $c = 25$ are squares and $2(3)(5) = 30 = b$, so it is a perfect square, $(3x + 5)^2$. $16x^2 - 24x + 9$: $a = 16$ and $c = 9$ are squares and $-2(4)(3) = -24 = b$, so it is a perfect square, $(4x - 3)^2$.

Lesson 8 — Extra Practice

Choosing a Factoring Strategy

Part A — Name the Strategy, Then Factor

Run the checklist from the top every time: **(1)** GCF first, **(2)** count the terms that are left, **(3)** check whether any factor can still be factored again. Write the name of the method (or the methods, in order) on the line, then carry it out.

1. Name the strategy, then factor completely. If an expression cannot be factored, write **prime**.

(a) $8x^2 - 18$
Strategy: _____

(b) $x^2 + 4$
Strategy: _____

(c) $4x^2 + 12x + 9$
Strategy: _____

(d) $3x^3 - 27x$
Strategy: _____

(e) $6x^2 - x - 15$
Strategy: _____

(f) $x^4 - 16$
Strategy: _____

2. Name the strategy, then factor completely. If an expression cannot be factored, write **prime**.

(a) $50 - 2x^2$
Strategy: _____

(b) $x^2 - x - 12$
Strategy: _____

(c) $9x^2 + 25$
Strategy: _____

(d) $12x^2 - 27$
Strategy: _____

(e) $x^2 - 20x + 100$
Strategy: _____

(f) $7x^3 + 14x^2 - 21x$
Strategy: _____

Part B — Two Stages and Three Stages

Step 3 of the checklist is the one people skip. After the GCF comes out, keep asking the same question of every factor you have written: *can this one be factored again?* Some expressions need two passes, and a few need three.

3. Factor completely. Each of these takes more than one step.

(a) $2x^4 - 32$

(b) $5x^3 + 10x^2 - 75x$

(c) $x^5 - x$

(d) $4x^3 - 4x^2 - 24x$

4. Factor completely.

(a) $2x^3 + 12x^2 + 18x$

(b) $18x^3 - 8x$

(c) $x^4 - 81$

(d) $6x^3 - 9x^2 - 6x$

Part C — Prime Polynomials, Impostors, and Unfinished Work

A polynomial is **prime** when no method on the checklist applies — most commonly a *sum* of squares such as $x^2 + 9$, or a trinomial whose factor pairs never give the right middle term. Saying "prime" is a conclusion, not a surrender: it needs a reason.

5. Factor completely, or write **prime** and give a one-line reason.

(a) $x^2 + 9$

(b) $x^2 - 9$

(c) $x^2 + 5x + 7$

(d) $x^2 + x - 6$

(e) $4x^2 + 9$

(f) $2x^2 + 8$

6. Each student below made a different mistake. Identify it, then give the correct factorization.

(a) Dev wrote $4x^2 - 36 = 4(x^2 - 9)$ and stopped.

(b) Mira wrote $x^2 + 25 = (x + 5)(x - 5)$.

7. Identify the mistake, then give the correct factorization.

(a) Jonah wrote $2x^2 + 10x + 12 = (2x + 4)(x + 3)$ and said he was finished.

(b) Priya wrote $x^4 - 16 = (x^2 - 4)(x^2 + 4)$ and said it was completely factored.

Part D — Sorting and Building

8. Sort the eight expressions below into the table by the **first** method you would use. Write each expression in exactly one row. Every row gets at least one expression, and exactly one row gets two.

$x^2 - 49$, $3x^2 + 6x$, $x^2 + 14x + 49$, $2x^2 + 5x - 3$, $5x^2 - 20$, $x^2 - 2x - 35$, $x^2 + 16$, $4x^3 - 4x$

First method	**Expressions**
GCF (and nothing after it)	
GCF, then another method	
Difference of squares	
Perfect square trinomial	
Trinomial, $a = 1$	
Trinomial, $a \neq 1$	
Prime — no method applies	

9. **Work backwards.** Build the expression that was factored, by multiplying the factors out. Write your answer in descending order with no brackets left.

(a) Build a three-term expression whose complete factorization is $3x(x + 4)(x - 1)$. Which two methods, in order, would a student need to get back to your answer?

(b) Build a two-term expression whose complete factorization is $2x(x - 5)(x + 5)$. Which two methods, in order, are needed?

10. **Build a prime one.** Each part asks for an expression that the checklist runs all the way through without ever factoring.
(a) Write a two-term expression in which both terms are perfect squares but the expression is still prime. Explain in one sentence why the difference-of-squares pattern does not reach it.

(b) Write a trinomial of the form $x^2 + bx + 10$ that is prime, and list every factor pair of 10 to show that no pair works.

Part E — Applications and Stretch

11. A rectangular box has volume $2x^3 - 8x$ cubic centimetres.
(a) Factor the volume completely to find three possible dimensions.

(b) State which dimension must be the largest when $x > 2$, and why.

(c) Determine the three dimensions when $x = 5$, then check that they multiply to the correct volume.

12. A storage bin has volume $3x^3 + 15x^2 + 18x$ cubic inches.
(a) Factor the volume completely.

(b) State the three dimensions.

(c) Explain why x cannot be 0 or a negative number here.

13. A square sheet of metal has side length $3x$ centimetres. A square hole of side length 12 centimetres is cut out of the middle.
(a) Write an expression for the remaining area, then factor it completely.

(b) Determine the remaining area when $x = 7$, using your factored form.

14. *(Multiple Choice)* $2x^4 - 32$ is completely factored as
A. $2(x^2 - 4)(x^2 + 4)$ B. $2(x - 2)(x + 2)(x^2 + 4)$
C. $(2x^2 - 8)(x^2 + 4)$ D. $2(x^2 - 4)^2$

15. *(Numerical Response)* If $4x^2 + kx + 25$ is a perfect square trinomial and $k > 0$, then the value of k is _____.

16. **Stretch — four linear factors.** A quartic such as $x^4 - 13x^2 + 36$ is a trinomial in disguise: treat x^2 as the variable, factor it as you would $n^2 - 13n + 36$, and then run step 3 on each factor you get.
(a) Factor $x^4 - 13x^2 + 36$ completely. Show both passes. (b) Factor $x^4 - 10x^2 + 9$ completely.

Answer Key. 1. (a) GCF, then difference of squares: $2(2x-3)(2x+3)$ (b) prime — a sum of squares never factors (c) perfect square trinomial: $(2x+3)^2$ (d) GCF, then difference of squares: $3x(x-3)(x+3)$ (e) trinomial, $a \neq 1$ (decomposition): $(2x+3)(3x-5)$ (f) difference of squares *twice*: $(x-2)(x+2)(x^2+4)$. **2.** (a) GCF, then difference of squares: $2(5-x)(5+x)$ (b) trinomial, $a = 1$ (inspection): $(x-4)(x+3)$ (c) prime — sum of squares (d) GCF, then difference of squares: $3(2x-3)(2x+3)$ (e) perfect square trinomial: $(x-10)^2$ (f) GCF, then trinomial $a = 1$: $7x(x+3)(x-1)$. **3.** (a) $2(x^2-4)(x^2+4) = 2(x-2)(x+2)(x^2+4)$ (b) $5x(x^2+2x-15) = 5x(x+5)(x-3)$ (c) $x(x^4-1) = x(x^2-1)(x^2+1) = x(x-1)(x+1)(x^2+1)$ (d) $4x(x^2-x-6) = 4x(x-3)(x+2)$. **4.** (a) $2x(x^2+6x+9) = 2x(x+3)^2$ (b) $2x(9x^2-4) = 2x(3x-2)(3x+2)$ (c) $(x^2-9)(x^2+9) = (x-3)(x+3)(x^2+9)$ (d) $3x(2x^2 - 3x - 2) = 3x(2x+1)(x-2)$. **5.** (a) prime (sum of squares) (b) $(x-3)(x+3)$ (c) prime — the only factor pair of 7 is 1×7, and $1 + 7 = 8 \neq 5$ (d) $(x+3)(x-2)$ (e) prime (sum of squares) (f) $2(x^2+4)$ — the GCF comes out, but x^2+4 is prime, so that is the complete answer. **6.** (a) He stopped one step early; $x^2 - 9$ is still a difference of squares, so $4x^2 - 36 = 4(x-3)(x+3)$. (b) She used the difference-of-squares pattern on a *sum*; $(x+5)(x-5) = x^2 - 25$, not $x^2 + 25$. In fact $x^2 + 25$ is prime. **7.** (a) He never removed the GCF, so a common factor of 2 is still trapped inside $(2x+4)$; correctly $2x^2 + 10x + 12 = 2(x^2 + 5x + 6) = 2(x+2)(x+3)$. (b) Her first pass is right but she stopped too early — $x^2 - 4$ is another difference of squares, giving $(x-2)(x+2)(x^2+4)$; x^2+4 is prime, so that is the end. **8.** GCF only: $3x^2 + 6x$. GCF then another method: $5x^2 - 20$, $4x^3 - 4x$. Difference of squares: $x^2 - 49$. Perfect square trinomial: $x^2 + 14x + 49$. Trinomial $a = 1$: $x^2 - 2x - 35$. Trinomial $a \neq 1$: $2x^2 + 5x - 3$. Prime: $x^2 + 16$. (Watch $5x^2 - 20$ and $4x^3 - 4x$: both look like a difference of squares, but the GCF always comes out first.) Factored forms: $3x(x+2)$, $5(x-2)(x+2)$, $4x(x-1)(x+1)$, $(x-7)(x+7)$, $(x+7)^2$, $(x-7)(x+5)$, $(2x-1)(x+3)$. **9.** (a) $3x^3 + 9x^2 - 12x$; GCF first, then a trinomial with $a = 1$. (b) $2x^3 - 50x$; GCF first, then difference of squares. **10.** (a) Sample: $x^2 + 36$ (or $4x^2 + 9$, $9x^2 + 1$). The pattern $a^2 - b^2$ needs a subtraction between the two squares; a sum of squares matches no pattern on the checklist. (b) Sample: $x^2 + 4x + 10$. The factor pairs of 10 are 1×10 and 2×5 (and their negatives), giving sums 11, 7, -11, -7; none is 4, so it is prime. **11.** (a) $2x^3 - 8x = 2x(x^2-4) = 2x(x-2)(x+2)$, so the dimensions are $2x$, $x-2$, $x+2$. (b) For $x > 2$, compare $2x$ with $x+2$: $2x > x+2$ exactly when $x > 2$, so $2x$ is the largest (and $x-2$ is always the smallest). (c) At $x = 5$: 10, 3, 7; $10 \times 3 \times 7 = 210$ and $2(125) - 8(5) = 250 - 40 = 210$. **12.** (a) $3x(x^2 + 5x + 6) = 3x(x+2)(x+3)$ (b) $3x$, $x+2$, $x+3$ (c) a dimension must be positive; $x = 0$ makes $3x = 0$ and any $x < 0$ makes $3x$ negative, so no bin exists. **13.** (a) $9x^2 - 144 = 9(x^2 - 16) = 9(x-4)(x+4)$ (b) $9(3)(11) = 297$ cm^2. **14.** B. **15.** 20 (since $4x^2 + 20x + 25 = (2x+5)^2$, and $k = 2 \times 2 \times 5$). **16.** (a) $x^4 - 13x^2 + 36 = (x^2-4)(x^2-9) = (x-2)(x+2)(x-3)(x+3)$ (product 36, sum -13, from -4 and -9). (b) $x^4 - 10x^2 + 9 = (x^2-1)(x^2-9) = (x-1)(x+1)(x-3)(x+3)$.

Lesson 9 — Extra Practice

Solving Polynomial Equations by Factoring

Part A — Rearranging So One Side Is Zero

The Zero Product Property says that if $A \cdot B = 0$, then $A = 0$ or $B = 0$. It says nothing at all about a product equal to 30, or to $8x$, or to anything else. So the very first move on every question below is to collect all terms on one side and leave a bare 0 on the other.

1. Rearrange each equation so one side is 0, then factor and solve.

(a) $x^2 + 7x = 30$

(b) $x^2 = 5x + 24$

(c) $x^2 - 9x = -20$

(d) $x^2 = 2x + 35$

2. Each equation below has a common factor. Remove the GCF first, then use the Zero Product Property. **Do not divide both sides by x.** State *both* roots every time.

(a) $3x^2 - 12x = 0$

(b) $2x^2 = 8x$

(c) $5x^2 + 35x = 0$

(d) $4x^2 = -20x$

3. **Error analysis.** Nadia solves $2x^2 = 8x$ like this: "Divide both sides by $2x$. That leaves $x = 4$." Her answer $x = 4$ is genuinely a solution, yet her work is still wrong. Explain exactly what her division destroyed, name the root she never found, and show the correct method.

Part B — Special Forms and a Leading Coefficient Other Than One

4. Solve each equation. Watch for a difference of squares and for a perfect square trinomial.

(a) $4x^2 = 25$

(b) $9x^2 - 64 = 0$

(c) $x^2 - 10x + 25 = 0$

(d) $x^2 + 14x + 49 = 0$

5. Parts (c) and (d) of question 4 each produced only *one* number, while every other equation on this page produced two. Explain why, using the factored form in your answer. A root of this kind has a name — state it.

6. Solve by decomposition. Every answer here is a fraction or involves one.

(a) $6x^2 - x - 2 = 0$

(b) $4x^2 + 4x - 15 = 0$

(c) $10x^2 + x - 3 = 0$

(d) $3x^2 = 10x - 8$

7. Rearrange, then solve by decomposition.

(a) $6x^2 + 5x = 6$

(b) $8x^2 = 2x + 15$

Part C — Expanding Before You Can Factor

An equation such as $(x+3)(x-2) = 6$ *looks* ready for the Zero Product Property, but it is not: the product equals 6, not 0. Expand the left side, move the 6 across, and only then factor the result. The factored form you finish with will not be the one you started with.

8. Expand, rearrange so one side is 0, then factor and solve.
(a) $(x+3)(x-2)=6$

(b) $(x-5)(x+1)=16$

(c) $x(x+5)=24$

(d) $(x+4)^2=25$

9. **Error analysis.** A student solves $(x+3)(x-2)=6$ by writing "$x+3=6$ or $x-2=6$," and reports $x=3$ and $x=8$.
(a) Substitute $x=3$ and then $x=8$ into the original equation. Which one actually works?

(b) The method is invalid even though one of its answers is correct. Explain why that answer arrived by luck, and state the correct root the student missed entirely.

10. Solve each equation. Expand first.
(a) $(2x+1)(x-3)=-6$

(b) $(x-1)(x+6)=8$

Part D — More Than Two Factors, More Than Two Roots

A polynomial equation of degree 3 can have up to three roots, and one of degree 4 up to four. The method never changes: make one side 0, factor *completely*, then set every factor containing the variable equal to 0. A GCF of x contributes the root $x=0$.

11. Solve each cubic equation. Each has three roots.
(a) $x^3-9x=0$

(b) $x^3+2x^2-15x=0$

(c) $2x^3-8x^2-24x=0$

(d) $3x(x-7)(x+2)=0$

12. Solve $(x-4)(x+1)(2x-3)=0$. Then explain why the factor $2x-3$ does not give the root $x=3$, even though the factor $x-4$ does give $x=4$.

13. **Stretch.** The equation $x^4-13x^2+36=0$ is a trinomial in x^2: think of it as $(x^2)^2-13(x^2)+36=0$. Factor it as a trinomial first, then factor each bracket again as a difference of squares, and finally solve. Expect four roots.
(a) $x^4-13x^2+36=0$ (b) $x^4-5x^2+4=0$

Part E — Checking, Building Backwards, and Applying

14. Solve $x^2+3x=28$ by factoring. Then substitute **each** root into the *original* equation and show the arithmetic that confirms it.
Solve: Check both roots:

15. **Work backwards.** If $x=r$ is a root, then $(x-r)$ is a factor. Use that idea to build a quadratic equation in the form $ax^2+bx+c=0$ with integer coefficients having the given roots. Expand your answer fully.
(a) roots -3 and 5 (b) roots 0 and 7 (c) roots $-\frac{1}{2}$ and 4

16. The product of two consecutive **positive** integers is 72. Let the smaller integer be n, write an equation, and solve it by factoring. Your equation will produce two roots — state which one you reject and why, then give the two integers.

17. The length of a rectangle is 3 cm more than its width, and its area is 40 cm^2. Let the width be w, write an equation, and solve by factoring. Reject the inadmissible root and give a reason, then state the dimensions.

18. The length of a rectangular deck is 4 m more than twice its width, and its area is 48 m^2. Determine the dimensions. State clearly which root you reject and why.

19. A ball is thrown from the top of a wall. Its height above the ground, in metres, after t seconds is given by $h = -5t^2 + 15t + 20$.

(a) How high was the ball when it was released? (Substitute $t = 0$.)

(b) Set $h = 0$ and solve by factoring to find when the ball lands. Reject one root and explain in words why it is impossible in this situation.

20. A rectangular garden measures 8 m by 5 m. A path of uniform width x metres is built all the way around the outside of it, and the garden and path together cover 154 m^2.

(a) Explain why the outer dimensions are $(8 + 2x)$ by $(5 + 2x)$, and write an equation for the total area.

(b) Solve by factoring, reject the inadmissible root with a reason, and state the width of the path.

Part F — Multiple Choice and Numerical Response

21. *(Multiple Choice)* The complete solution to the equation $(x - 1)(x + 4) = 6$ is A. $x = 1$ and $x = -4$ B. $x = 7$ and $x = 2$ C. $x = -5$ and $x = 2$ D. $x = 5$ and $x = -2$

22. *(Numerical Response)* The equation $x^2 - 11x + k = 0$ has one solution $x = 3$. The value of k is _____.

23. *(Numerical Response)* The non-zero solution of $4x^2 - 9x = 0$ can be written as $x = \frac{p}{q}$ in lowest terms, where p and q are positive integers. The value of $p + q$ is _____.

Answer Key. 1. (a) $x^2 + 7x - 30 = 0$, $(x + 10)(x - 3) = 0$, $x = -10, 3$ (b) $x^2 - 5x - 24 = 0$, $(x - 8)(x + 3) = 0$, $x = 8, -3$ (c) $x^2 - 9x + 20 = 0$, $(x - 4)(x - 5) = 0$, $x = 4, 5$ (d) $x^2 - 2x - 35 = 0$, $(x - 7)(x + 5) = 0$, $x = 7, -5$. **2.** (a) $3x(x - 4) = 0$, $x = 0, 4$ (b) $2x^2 - 8x = 0$, $2x(x - 4) = 0$, $x = 0, 4$ (c) $5x(x + 7) = 0$, $x = 0, -7$ (d) $4x^2 + 20x = 0$, $4x(x + 5) = 0$, $x = 0, -5$. **3.** Dividing by $2x$ is only legal if $2x \neq 0$, and dividing throws that case away. The lost root is $x = 0$ (check: $2(0)^2 = 0$ and $8(0) = 0$, so it truly solves the equation). Correctly: $2x^2 - 8x = 0$, $2x(x - 4) = 0$, so $x = 0$ or $x = 4$. Never divide both sides by a variable — subtract instead, and factor. **4.** (a) $4x^2 - 25 = 0$, $(2x - 5)(2x + 5) = 0$, $x = \pm\frac{5}{2}$ (b) $(3x - 8)(3x + 8) = 0$, $x = \pm\frac{8}{3}$ (c) $(x - 5)^2 = 0$, $x = 5$ only (d) $(x + 7)^2 = 0$, $x = -7$ only. **5.** A perfect square trinomial factors into two *identical* factors, so both small equations are the same one: $(x - 5)(x - 5) = 0$ gives $x - 5 = 0$ twice. There is only one distinct value. It is called a **double root** (or repeated root). **6.** (a) $(2x + 1)(3x - 2) = 0$, $x = -\frac{1}{2}, \frac{2}{3}$ (b) $(2x - 3)(2x + 5) = 0$, $x = \frac{3}{2}, -\frac{5}{2}$ (c) $(2x - 1)(5x + 3) = 0$, $x = \frac{1}{2}, -\frac{3}{5}$ (d) $3x^2 - 10x + 8 = 0$, $(x - 2)(3x - 4) = 0$, $x = 2, \frac{4}{3}$. **7.** (a) $6x^2 + 5x - 6 = 0$, $(2x + 3)(3x - 2) = 0$, $x = -\frac{3}{2}, \frac{2}{3}$ (b) $8x^2 - 2x - 15 = 0$, $(2x - 3)(4x + 5) = 0$, $x = \frac{3}{2}, -\frac{5}{4}$. **8.** (a) $x^2 + x - 6 = 6$, so $x^2 + x - 12 = 0$, $(x + 4)(x - 3) = 0$, $x = -4, 3$ (b) $x^2 - 4x - 5 = 16$, so $x^2 - 4x - 21 = 0$, $(x - 7)(x + 3) = 0$, $x = 7, -3$ (c) $x^2 + 5x - 24 = 0$, $(x + 8)(x - 3) = 0$, $x = -8, 3$ (d) $x^2 + 8x + 16 = 25$, so $x^2 + 8x - 9 = 0$, $(x + 9)(x - 1) = 0$, $x = -9, 1$. **9.** (a) $x = 3$: $(3 + 3)(3 - 2) = 6 \times 1 = 6$ ✓. $x = 8$: $(8 + 3)(8 - 2) = 11 \times 6 = 66 \neq 6$, which fails. Only $x = 3$ works. (b) The method assumes one factor must equal 6, which is false — 6 can be split as 6×1, 2×3, $(-2) \times (-3)$, $12 \times \frac{1}{2}$, and endlessly more, whereas only 0 forces a factor to be 0. The student's $x = 3$ happens to make the split 6×1 occur, so it is a coincidence, not a consequence. The correct work in 8(a) gives $x = -4$ and $x = 3$; the root $x = -4$ was missed completely. **10.** (a) $2x^2 - 5x - 3 = -6$, so $2x^2 - 5x + 3 = 0$, $(x - 1)(2x - 3) = 0$, $x = 1, \frac{3}{2}$ (b) $x^2 + 5x - 6 = 8$, so $x^2 + 5x - 14 = 0$, $(x + 7)(x - 2) = 0$, $x = -7, 2$. **11.** (a) $x(x - 3)(x + 3) = 0$, $x = 0, 3, -3$ (b) $x(x + 5)(x - 3) = 0$, $x = 0, -5, 3$ (c) $2x(x - 6)(x + 2) = 0$, $x = 0, 6, -2$ (d) $x = 0, 7, -2$ (the constant 3 gives no root). **12.** $x = 4$, $x = -1$, $x = \frac{3}{2}$. Setting $2x - 3 = 0$ gives $2x = 3$, so $x = \frac{3}{2}$ — the coefficient 2 must be divided out at the end. Only a factor of the form $x - r$ hands you the root directly. **13.** (a) $(x^2 - 4)(x^2 - 9) = 0$, then $(x - 2)(x + 2)(x - 3)(x + 3) = 0$, so $x = \pm 2, \pm 3$ (b) $(x^2 - 1)(x^2 - 4) = 0$, then $(x - 1)(x + 1)(x - 2)(x + 2) = 0$, so $x = \pm 1, \pm 2$. **14.** $x^2 + 3x - 28 = 0$, $(x + 7)(x - 4) = 0$, $x = -7, 4$. Check $x = -7$: $(-7)^2 + 3(-7) = 49 - 21 = 28$ ✓. Check $x = 4$: $(4)^2 + 3(4) = 16 + 12 = 28$ ✓. **15.** (a) $(x + 3)(x - 5) = 0$, so $x^2 - 2x - 15 = 0$ (b) $x(x - 7) = 0$, so $x^2 - 7x = 0$ (c) $(2x + 1)(x - 4) = 0$, so $2x^2 - 7x - 4 = 0$ (the factor $2x + 1$ clears the fraction). **16.** $n(n + 1) = 72$, so $n^2 + n - 72 = 0$, $(n + 9)(n - 8) = 0$, $n = -9$ or $n = 8$. Reject $n = -9$: the question asks for *positive* integers, and -9 is negative. So the integers are 8 and 9 ($8 \times 9 = 72$ ✓). **17.** $w(w + 3) = 40$, so $w^2 + 3w - 40 = 0$, $(w + 8)(w - 5) = 0$, $w = -8$ or $w = 5$. Reject $w = -8$: a width is a physical length and cannot be negative. Width 5 cm, length 8 cm; $5 \times 8 = 40$ ✓. **18.** $w(2w + 4) = 48$, so $2w^2 + 4w - 48 = 0$, $2(w + 6)(w - 4) = 0$, $w = -6$ or $w = 4$. Reject $w = -6$ (a width cannot be negative). Width 4 m, length $2(4) + 4 = 12$ m; $4 \times 12 = 48$ ✓. **19.** (a) $h = -5(0)^2 + 15(0) + 20 = 20$ m. (b) $-5t^2 + 15t + 20 = 0$, divide by -5 to get $t^2 - 3t - 4 = 0$, $(t - 4)(t + 1) = 0$, $t = 4$ or $t = -1$. Reject $t = -1$: time is measured forward from the moment of release, so a negative time is before the ball was thrown and has no meaning here. The ball lands after 4 s (check: $-5(16) + 15(4) + 20 = -80 + 60 + 20 = 0$ ✓). **20.** (a) The path adds x metres at *each* end of both dimensions, so each dimension grows by $2x$: $(8 + 2x)(5 + 2x) = 154$. (b) Expanding gives $4x^2 + 26x + 40 = 154$, so $4x^2 + 26x - 114 = 0$, or $2x^2 + 13x - 57 = 0$, which factors as $(2x + 19)(x - 3) = 0$, giving $x = -\frac{19}{2}$ or $x = 3$. Reject $x = -\frac{19}{2}$: a path width cannot be negative. The path is 3 m wide (check: $14 \times 11 = 154$ ✓). **21.** C. ($x^2 + 3x - 4 = 6$, so $x^2 + 3x - 10 = 0$, $(x + 5)(x - 2) = 0$.) **22.** 24. ($(3)^2 - 11(3) + k = 0$ gives $9 - 33 + k = 0$, so $k = 24$; the other root is 8.) **23.** 13. ($x(4x - 9) = 0$, so $x = 0$ or $x = \frac{9}{4}$; $p + q = 9 + 4 = 13$.)

Lesson 10 — Extra Practice

Unit Review: Factoring

Part A — Factor Completely: More Than One Stage

Every expression in this part needs **two or three** steps. Remove the greatest common factor first, then keep checking each bracket you write — the job is not finished until nothing left can be factored.

1. Factor completely.

(a) $4x^4 - 64$

(b) $3x^3 - 27x$

(c) $2x^3 + 10x^2 + 12x$

(d) $5x^3 - 45x$

2. Factor completely. In each one the GCF hides a difference of squares, a perfect square trinomial, or a trinomial with $a = 1$.

(a) $18x^3 - 8x$

(b) $2x^4 - 32$

(c) $3x^3 + 21x^2 + 30x$

(d) $4x^3 - 24x^2 + 36x$

3. Factor completely. After the GCF comes out, each of these still has $a \neq 1$, so finish with decomposition. In part (d), factor out a **negative** GCF.

(a) $12x^2 + 2x - 4$

(b) $8x^3 - 2x^2 - 3x$

(c) $6x^3 + 9x^2 - 6x$

(d) $-2x^2 + 16x - 32$

Part B — Recognising the Structure

4. The six expressions below are built from the same few numbers, but each one has a different structure. For each, **name the strategy** (GCF, difference of squares, perfect square trinomial, sum–product, decomposition, or none) and then factor it. Two of the six are **prime** — say so and explain why.

(a) $4x^2 - 25$ (b) $4x^2 + 25$ (c) $4x^2 - 20x + 25$

(d) $4x^2 + 20x + 25$ (e) $4x^2 + 25x + 25$ (f) $4x^2 + 5x + 25$

5. Name the strategy, then factor completely. One of these is prime.

(a) $7x^2 - 63$ (b) $x^2 - 7x - 18$ (c) $9x^2 - 30x + 25$

(d) $x^2 + 13x + 36$ (e) $5x^2 + 3x$ (f) $x^2 + 4x + 6$

6. Each expression below *looks* like a perfect square trinomial. Decide whether it really is one. If it is, write it as a square; if it is not, factor it completely by another method.

(a) $x^2 + 14x + 49$ (b) $x^2 + 14x + 48$ (c) $9x^2 - 24x + 16$ (d) $9x^2 - 24x + 15$

Part C — Solving Equations

Not one of these equations is ready to solve as written. Rearrange so that one side is 0, factor completely, then apply the Zero Product Property. Check every root by substituting it back into the *original* equation.

7. Solve.

(a) $(x + 4)(x - 1) = 6$ (b) $6x^2 = x + 2$

8. Solve. Each of these is a **cubic**, so expect three roots (or three factors).

(a) $x^3 = 4x$ (b) $2x^3 + 2x^2 = 12x$

9. Solve. One of these has only a single solution — when you find it, say why.
(a) $3x^2 + 27 = 18x$ (b) $x(x + 5) = 14$

10. Solve.
(a) $(x + 2)(x + 6) = 12$ (b) $2x(x - 3) = x^2 - 9$

Part D — Reasoning and Error Analysis

11. Both binomials below use the numbers x^2 and 4.
(a) Factor $x^2 - 4$, then explain in your own words why $x^2 + 4$ cannot be factored over the integers. Use the sum–product reasoning: what would $p + q$ and pq have to be?
(b) Use the same reasoning to explain why $x^2 + 4x + 6$ is prime. List every pair of integers whose product is 6 and show that none of them adds to 4.

12. A quadratic equation usually has two different solutions, but sometimes it has only one — a **repeated root**.
(a) Solve $x^2 - 6x + 9 = 0$. How many different roots are there, and what is it about the factored form that tells you?
(b) Complete the sentence: a quadratic has a repeated root exactly when its factored form is _____.
(c) Which of $x^2 - 10x + 25$, $x^2 - 10x + 24$, and $4x^2 + 12x + 9$ give equations with a repeated root?

13. When an expression is factored completely and one of the factors is simply x, the equation *expression* $= 0$ always has $x = 0$ as one of its roots.
(a) Explain why, using the Zero Product Property. Then state, without doing any more work, one root of $2x^3 + 2x^2 - 12x = 0$.
(b) The equation $5x^2 + 3x = 0$ has a common factor of x. Solve it, and say which root came from the GCF and which came from the other factor.

14. Two students made exactly **one** mistake each. Find the step where each went wrong, explain what the mistake was, and give the correct final answer.

(a) Rowan was asked to factor $3x^4 - 48$ completely:
Step 1: $3x^4 - 48 = 3(x^4 - 16)$
Step 2: $= 3(x^2 - 4)(x^2 + 4)$
Step 3: "This is factored completely."
Which step is wrong, and what is the complete factorization?

(b) Maya was asked to solve $x^2 = 5x$:
Step 1: $x^2 = 5x$
Step 2: divide both sides by x
Step 3: $x = 5$
Which step is wrong, what did Maya lose, and what are *all* the solutions?

Part E — Working Backwards

15. Build an equation, in the form *(polynomial)* $= 0$ with integer coefficients and no common factor, whose roots are exactly the numbers given. Expand your answer.

(a) $x = 3$ and $x = -5$

(b) $x = \frac{1}{2}$ and $x = -4$

(c) $x = 0$ and $x = 7$

16. Missing coefficients.

(a) Determine **both** values of k that make $x^2 + kx + 25$ a perfect square trinomial, and write the square for each.

(b) List every natural number c for which $x^2 + 10x + c$ factors over the integers.

(c) Which value of c from part (b) gives a perfect square trinomial?

17. Determine **every** integer value of b for which $2x^2 + bx + 3$ can be factored over the integers. Show the factored form for each, and explain how you know your list is complete.

Part F — Applications

18. The area of a rectangle is $(2x^2 + 7x + 3)$ cm^2.

(a) Determine expressions for the length and the width.

(b) Determine a simplified expression for the perimeter.

(c) Determine the actual dimensions, perimeter, and area when $x = 4$.

19. The volume of a rectangular box is $(x^3 + 6x^2 + 8x)$ cm^3.

(a) Factor the volume completely to find expressions for the three dimensions.

(b) Determine the three dimensions when $x = 3$, and verify that they multiply to the correct volume.

(c) Explain why x must be greater than 0 for this box to exist.

20. Consecutive-integer problems.

(a) The product of two consecutive positive integers is 132. Let the smaller be n, write an equation, and solve it by factoring. State both roots and say why one is rejected.

(b) The product of two consecutive *even* integers is 168. Let the smaller be n, write an equation, and solve it by factoring. Give both pairs of integers that work.

21. The legs of a right triangle measure x cm and $(x+7)$ cm, and the hypotenuse measures $(x+8)$ cm. Use the Pythagorean theorem to write an equation, simplify it to a quadratic, solve by factoring, reject the impossible root, and state the three side lengths.

Part G — Challenge

22. Factor $x^4 - 5x^2 + 4$ completely. *Hint:* treat x^2 as the variable and factor the trinomial first; then look hard at each bracket. Your final answer should be a product of **four** linear factors.

23. Solve $x^4 - 10x^2 + 9 = 0$. Factor completely first, then use the Zero Product Property. How many solutions are there, and why can a fourth-degree equation have that many?

24. Factor $(x+3)^2 - 16$ completely. *Hint:* treat the bracket as one square: $A^2 - B^2 = (A-B)(A+B)$ with $A = x+3$ and $B = 4$. Then simplify each bracket, and check your answer by expanding both $(x+3)^2 - 16$ and your factored form.

Answer Key. 1. (a) $4(x-2)(x+2)(x^2+4)$ (b) $3x(x-3)(x+3)$ (c) $2x(x+2)(x+3)$ (d) $5x(x-3)(x+3)$. **2.** (a) $2x(3x-2)(3x+2)$ (b) $2(x-2)(x+2)(x^2+4)$ (c) $3x(x+2)(x+5)$ (d) $4x(x-3)^2$. **3.** (a) $2(2x-1)(3x+2)$ (b) $x(4x-3)(2x+1)$ (c) $3x(2x-1)(x+2)$ (d) $-2(x-4)^2$. **4.** (a) difference of squares, $(2x-5)(2x+5)$ (b) **prime** — a *sum* of squares never factors over the integers (c) perfect square trinomial, $(2x-5)^2$ (d) perfect square trinomial, $(2x+5)^2$ (e) decomposition, $(4x+5)(x+5)$ (f) **prime** — $ac = 100$ and no two integers with product 100 add to 5. **5.** (a) GCF then difference of squares, $7(x-3)(x+3)$ (b) sum–product, $(x-9)(x+2)$ (c) perfect square trinomial, $(3x-5)^2$ (d) sum–product, $(x+4)(x+9)$ (e) GCF only, $x(5x+3)$ (f) prime — no integer pair has product 6 and sum 4. **6.** (a) yes, $(x+7)^2$ (b) no (48 is not a perfect square), $(x+6)(x+8)$ (c) yes, $(3x-4)^2$ (d) no (15 is not a perfect square), $3(3x-5)(x-1)$. **7.** (a) $x^2+3x-10=0$, $(x+5)(x-2)=0$, $x=-5$ or $x=2$ (b) $6x^2-x-2=0$, $(3x-2)(2x+1)=0$, $x=\frac{2}{3}$ or $x=-\frac{1}{2}$. **8.** (a) $x^3-4x=0$, $x(x-2)(x+2)=0$, $x=0,\ 2,\ -2$ (b) $2x^3+2x^2-12x=0$, $2x(x+3)(x-2)=0$, $x=0,\ -3,\ 2$. **9.** (a) $3x^2-18x+27=0$, $3(x-3)^2=0$, $x=3$ only — the same factor appears twice, so there is one repeated root (b) $x^2+5x-14=0$, $(x+7)(x-2)=0$, $x=-7$ or $x=2$. **10.** (a) $x^2+8x+12=12$, so $x^2+8x=0$, $x(x+8)=0$, $x=0$ or $x=-8$ (b) $2x^2-6x=x^2-9$, so $x^2-6x+9=0$, $(x-3)^2=0$, $x=3$ (repeated). **11.** (a) $x^2-4=(x-2)(x+2)$. For $x^2+4=(x+p)(x+q)$ we would need $p+q=0$ and $pq=4$; $p+q=0$ forces $q=-p$, so $pq=-p^2$, which is never positive. No integers work, so x^2+4 is prime. (b) Pairs with product 6: 1 and 6 (sum 7), 2 and 3 (sum 5), -1 and -6 (sum -7), -2 and -3 (sum -5). None sums to 4, so x^2+4x+6 is prime. **12.** (a) $(x-3)^2=0$, so $x=3$ — one root, because the *same* factor appears twice (b) a perfect square, $a(px+q)^2$ — that is, both factors are identical (c) $x^2-10x+25=(x-5)^2$ and $4x^2+12x+9=(2x+3)^2$ have repeated roots; $x^2-10x+24=(x-4)(x-6)$ has two different roots. **13.** (a) If the factored form is $x\cdot(\text{rest})=0$, then one of the factors is x itself, and $x=0$ makes that factor zero, so $x=0$ is a root. Therefore $x=0$ is a root of $2x^3+2x^2-12x=0$. (b) $x(5x+3)=0$, so $x=0$ (from the GCF x) or $x=-\frac{3}{5}$ (from $5x+3=0$). **14.** (a) Step 3 is wrong — Rowan stopped too early. x^2-4 is still a difference of squares, so $3x^4-48=3(x-2)(x+2)(x^2+4)$. Only x^2+4 (a sum of squares) is finished. (b) Step 2 is wrong — dividing by x is only legal if $x\neq 0$, and here it throws away the root $x=0$. Instead move everything to one side: $x^2-5x=0$, $x(x-5)=0$, so $x=0$ or $x=5$. **15.** (a) $(x-3)(x+5)=0$, i.e. $x^2+2x-15=0$ (b) $(2x-1)(x+4)=0$, i.e. $2x^2+7x-4=0$ (c) $x(x-7)=0$, i.e. $x^2-7x=0$. **16.** (a) $k=10$ gives $(x+5)^2$ and $k=-10$ gives $(x-5)^2$ (the middle term must be $\pm 2(1)(5)$) (b) $c=9,\ 16,\ 21,\ 24,\ 25$, from the pairs 1+9, 2+8, 3+7, 4+6, 5+5 (c) $c=25$, giving $(x+5)^2$. **17.** $b=5,\ 7,\ -5,\ -7$: $2x^2+5x+3=(2x+3)(x+1)$, $2x^2+7x+3=(2x+1)(x+3)$, $2x^2-5x+3=(2x-3)(x-1)$, $2x^2-7x+3=(2x-1)(x-3)$. The list is complete because the factors must be $(2x+m)(x+n)$ with $mn=3$, and the only integer pairs are $(1,3)$, $(3,1)$, $(-1,-3)$, $(-3,-1)$, giving $b=m+2n=7,\ 5,\ -7,\ -5$. **18.** (a) $2x^2+7x+3=(2x+1)(x+3)$, so the sides are $(2x+1)$ cm and $(x+3)$ cm (b) $P=2\big[(2x+1)+(x+3)\big]=(6x+8)$ cm (c) at $x=4$: 9 cm by 7 cm, $P=32$ cm, $A=63$ cm^2 (and $2(4)^2+7(4)+3=63$, which checks). **19.** (a) $x^3+6x^2+8x=x(x+2)(x+4)$, so the dimensions are x, $(x+2)$, $(x+4)$ cm (b) at $x=3$: 3 cm, 5 cm, 7 cm, and $3\times 5\times 7=105$ cm^3, matching $27+54+24=105$ (c) if $x\leq 0$ then the dimension x would be zero or negative, and a length must be positive. **20.** (a) $n(n+1)=132$, so $n^2+n-132=0$, $(n-11)(n+12)=0$, $n=11$ or $n=-12$; reject $n=-12$ because the integers must be positive, so the integers are 11 and 12 (b) $n(n+2)=168$, so $n^2+2n-168=0$, $(n-12)(n+14)=0$, $n=12$ or $n=-14$; both give even pairs — 12 and 14, or -14 and -12. **21.** $x^2+(x+7)^2=(x+8)^2$ becomes $2x^2+14x+49=x^2+16x+64$, so $x^2-2x-15=0$, $(x-5)(x+3)=0$, $x=5$ or $x=-3$; reject $x=-3$ (a length cannot be negative). The sides are 5 cm, 12 cm, and 13 cm. **22.** $x^4-5x^2+4=(x^2-1)(x^2-4)=(x-1)(x+1)(x-2)(x+2)$ — each bracket is itself a difference of squares. **23.** $x^4-10x^2+9=(x^2-1)(x^2-9)=(x-1)(x+1)(x-3)(x+3)=0$, so $x=1,\ -1,\ 3,\ -3$ — four solutions, because a degree-four polynomial can have up to four linear factors, and each factor contributes one root. **24.** $(x+3)^2-16=\big((x+3)-4\big)\big((x+3)+4\big)=(x-1)(x+7)$. Check: $(x+3)^2-16=x^2+6x+9-16=x^2+6x-7$ and $(x-1)(x+7)=x^2+6x-7$ ✓.